

spicyGLM for Spatial Co-localisation

Derivation, Notation, Closed-Form Simplifications, Inference, and Diagnostics

Contents

1	Model Setup	4
1.1	Indexing	4
1.2	Mean Model	4
2	Notation	5
3	The Sandwich Estimator	7
3.1	The Information Matrix and Bread	7
3.2	The Meat	7
3.3	The Sandwich	7
4	The CR2 Adjustment	8
4.1	Why the Plain Sandwich is Biased	8
4.2	The BRL Criterion	8
4.3	Solution via the Inner Matrix G_i	9
4.4	The CR2 Sandwich Estimator	9
4.5	Simplification of G_i for the Poisson GLM	10
5	Closed-Form $G_i^{-1/2}$ Under One Image Per Patient	10
5.1	Matrix Structure Under One Image Per Patient	10
5.2	Information Matrix and Bread	11
5.3	Leverage Scalar α_i	11
5.4	Structure of G_i	12
5.5	Eigenvalues of G_i	12
5.6	Closed-Form $G_i^{-1/2}$	12
5.7	The CR2 Adjustment Matrix A_i	13
5.8	Computational Algorithm	13
6	Closed-Form $G_i^{-1/2}$ Under Multiple Images Per Patient	13
6.1	Matrix Structure Under Multiple Images Per Patient	13
6.2	The Image-Indicator Subspace and Its Complement	14
6.3	Structure of G_i : The DPR1 Form	15

6.4	Closed-Form Eigenpairs from the Within-Image Complement	16
6.5	The Reduced Eigenvalue Problem on the Image-Indicator Subspace	16
6.6	Solving the Reduced Problem via DPR1 Theory	17
6.7	Closed-Form Assembly of $G_i^{-1/2}$	19
6.8	The CR2 Adjustment Matrix A_i (General Case)	20
6.9	Complexity	22
6.10	Computational Algorithm	22
6.11	The <code>dpr1eig</code> Algorithm, Adapted to Our Notation	23
6.12	Solving the Reduced Problem via Direct Eigendecomposition	24
7	Implementation Plan (Overview)	25
8	Empirical Comparison of Eigendecomposition Routes	25
8.1	Method	26
8.2	Results	26
9	Degrees of Freedom	26
9.1	Setup	27
9.2	Sensitivity of the Estimating Equation	27
9.3	The Loading Vector g_j	27
9.4	Exact Distribution of $\hat{v}$	28
9.5	Closed Form of P_{jk}	30
9.6	Computational Algorithm	31
10	Firth Correction	32
10.1	Motivation: Boundary Failure of the MLE	32
10.2	Firth's Penalized Likelihood: General Derivation	32
10.3	Closed-Form Firth Estimate for <code>spicyGLM</code>	34
10.4	Comparison to the Crude Per-Cell Correction	35
11	Diagnostics	35
11.1	Motivation	35
11.2	Leverage	36
11.2.1	Patient-Level Leverage	36
11.2.2	Image-Level Decomposition	36
11.3	Influence	37
11.3.1	Motivation	37
11.3.2	Derivation	37
11.3.3	Definition	37
11.3.4	Relationship to Leave-One-Out Influence	38
11.3.5	Image-Level Decomposition	38
11.4	Point-Estimate Shift	38
11.5	Reading the Diagnostics Together	39
11.6	Cross-Pair Aggregation	39
11.6.1	The Comparability Problem	39

11.6.2	Relative Leverage, Influence, and Point-Estimate Shift	40
11.6.3	Within-Pair Percentile Normalisation	40
11.6.4	Cross-Pair Flagging	41
11.6.5	Image-Level Mean-Share Summaries	41
12	Extension to the Binomial GLM (Fixed-k Nearest-Neighbour Design)	43
12.1	Motivation and Model Setup	43
12.2	Notation Changes	44
12.3	The CR2 Adjustment and Degrees of Freedom Survive Unchanged	45
12.4	Firth's Correction Does Not Admit a Closed Form	47

1 Model Setup

1.1 Indexing

Let m denote the number of patients. For patient $i = 1, \dots, m$, let n_i denote the number of images. For image $j = 1, \dots, n_i$ of patient i , let n_{ij} denote the number of reference cells. The total number of reference cells for patient i is:

$$N_i = \sum_{j=1}^{n_i} n_{ij}$$

and the total number of reference cells across all patients is:

$$N = \sum_{i=1}^m N_i.$$

1.2 Mean Model

spicyGLM models spatial co-localisation between a reference cell type (REF) and a target cell type (TARGET). For each REF cell, the response is the number of TARGET neighbours within a fixed radius r .

Patients are partitioned into two groups, indexed by $g(i) \in \{1, 2\}$. The model is a no-intercept Poisson GLM with $p = 2$ parameters:

$$\beta = (\beta_1, \beta_2)^\top \in \mathbb{R}^2,$$

where β_g is the log expected neighbour count for a REF cell in group g , after adjusting for image-level target density.

For cell $k = 1, \dots, n_{ij}$ in image j of patient i , the model is:

$$Y_{ijk} \sim \text{Poisson}(\mu_{ijk}), \quad \log \mu_{ijk} = \beta_{g(i)} + \log(\widehat{\text{den}}_{ij}),$$

where $\log(\widehat{\text{den}}_{ij})$ is an image-level offset encoding the expected number of TARGET cells within radius r under spatial randomness.

The covariate vector for cell k in image j of patient i is the standard basis vector in $\mathbb{R}^2$ selecting group $g(i)$:

$$x_{ijk} = \mathbf{e}_{g(i)} = \begin{cases} (1, 0)^\top & \text{if patient } i \text{ belongs to group 1,} \\ (0, 1)^\top & \text{if patient } i \text{ belongs to group 2.} \end{cases}$$

This encodes group membership directly in the covariate, so that $x_{ijk}^\top \beta = \beta_{g(i)}$ — selecting the appropriate group-level log mean. All cells within a patient share the same covariate vector since group membership is a patient-level attribute.

Interpretation of β . The exponentiated coefficients $\exp(\beta_g)$ represent the population-average multiplicative enrichment in TARGET neighbours for a REF cell in group g , relative to the image-level density baseline:

$$\exp(\beta_g) > 1 \implies \text{spatial enrichment (attraction)}, \quad \exp(\beta_g) < 1 \implies \text{spatial depletion (avoidance)}.$$

The primary scientific contrast is $\beta_1 - \beta_2$, the log fold-change in enrichment between the two groups.

2 Notation

For each patient i , we stack all observations across all images into the following vectors and matrices. We also define the full $N \times N$ hat matrix and its cluster-row submatrices, which are needed for the CR2 adjustment in Section 4.

Symbol	Dimension	Description
<i>Indexing</i>		
m	scalar	number of patients
n_i	scalar	number of images for patient i
n_{ij}	scalar	number of REF cells in image j of patient i
$N_i = \sum_{j=1}^{n_i} n_{ij}$	scalar	total REF cells for patient i
$N = \sum_{i=1}^m N_i$	scalar	total REF cells overall
$g(i) \in \{1, 2\}$	scalar	group membership of patient i
<i>Model objects</i>		
Y_{ijk}	scalar	TARGET neighbour count for cell k , image j , patient i
μ_{ijk}	scalar	expected neighbour count
$x_{ijk} = \mathbf{e}_{g(i)}$	2×1	covariate vector (standard basis vector for group $g(i)$)
$\beta = (\beta_1, \beta_2)^\top$	2×1	log expected neighbour counts per group
<i>Stacked objects for patient i</i>		
Y_i	$N_i \times 1$	response vector

Symbol	Dimension	Description
$X_i = \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top$	$N_i \times 2$	design matrix (all rows equal $\mathbf{e}_{g(i)}^\top$)
μ_i	$N_i \times 1$	mean vector
$\hat{r}_i = Y_i - \hat{\mu}_i$	$N_i \times 1$	fitted residuals
$V_i = \text{diag}(\mu_i)$	$N_i \times N_i$	working variance matrix
$W_i = V_i^{-1} = \text{diag}(1/\mu_i)$	$N_i \times N_i$	inverse variance matrix
$D_i = V_i^{1/2} = \text{diag}(\sqrt{\mu_i})$	$N_i \times N_i$	symmetric square root of V_i
<i>Full-data stacked objects</i>		
$X = (X_1^\top, \dots, X_m^\top)^\top$	$N \times 2$	full design matrix
$\mathbf{V} = \text{blkdiag}(V_1, \dots, V_m)$	$N \times N$	block-diagonal working variance
<i>Sandwich and CR2 objects</i>		
$\mathcal{I} = \sum_{i=1}^m X_i^\top V_i X_i$	2×2	Fisher information matrix
$\mathcal{B} = \mathcal{I}^{-1}$	2×2	bread
$H_X = \mathbf{V} X \mathcal{B} X^\top$	$N \times N$	full hat matrix
$(I - H_X)_i$	$N_i \times N$	rows of $(I - H_X)$ belonging to patient i ; an $N_i \times N$ matrix, <i>not</i> the $N_i \times N_i$ diagonal block
$H_{X,ii} = V_i X_i \mathcal{B} X_i^\top$	$N_i \times N_i$	diagonal block of H_X for patient i
$Q_i = (I - H_X)_i \mathbf{V} (I - H_X)_i^\top$	$N_i \times N_i$	first-order residual covariance for patient i
$G_i = D_i Q_i D_i$	$N_i \times N_i$	inner matrix for CR2
$A_i = D_i G_i^{-1/2} D_i$	$N_i \times N_i$	CR2 adjustment matrix
$\mathcal{M} = \sum_{i=1}^m X_i^\top \hat{r}_i \hat{r}_i^\top X_i$	2×2	unadjusted meat (CR0)
$\widetilde{\mathcal{M}} = \sum_{i=1}^m X_i^\top A_i \hat{r}_i \hat{r}_i^\top A_i^\top X_i$	2×2	CR2-adjusted meat
$\hat{V}_{\text{CR2}} = \mathcal{B} \widetilde{\mathcal{M}} \mathcal{B}$	2×2	CR2 sandwich estimator
<i>One-image-per-patient objects</i>		
$S_g = \sum_{i: g(i)=g} N_i \hat{\mu}_i$	scalar	total weighted cell count in group g
$\alpha_i = \hat{\mu}_i / S_{g(i)}$	scalar	leverage scalar for patient i

Table 2: Notation for spicyGLM and the CR2 sandwich estimator. Note that $(I - H_X)_i$ denotes the N_i rows of the full $N \times N$ residual projection matrix $(I - H_X)$ belonging to patient i ; it is an $N_i \times N$ matrix, not the $N_i \times N_i$ diagonal block $I_{N_i} - H_{X,ii}$.

3 The Sandwich Estimator

3.1 The Information Matrix and Bread

The Poisson GLM score equation is:

$$U(\beta) = \sum_{i=1}^m X_i^\top (Y_i - \mu_i) = 0.$$

The Fisher information matrix is the negative expected Jacobian of $U(\beta)$:

$$\mathcal{I} = \sum_{i=1}^m X_i^\top V_i X_i \in R^{2 \times 2},$$

where $V_i = \text{diag}(\mu_i)$ is the working variance matrix for patient i . Under a correctly specified Poisson GLM with independent observations, the asymptotic covariance of $\hat{\beta}$ is:

$$\text{Var}(\hat{\beta}) = \mathcal{B} = \mathcal{I}^{-1}.$$

This is the *model-based* variance estimator. However, in spicyGLM observations within a patient are correlated, so $\mathcal{B}$ alone underestimates the true variance of $\hat{\beta}$.

3.2 The Meat

The meat captures the *empirical* variability in the score contributions across patients:

$$\mathcal{M} = \sum_{i=1}^m X_i^\top \hat{r}_i \hat{r}_i^\top X_i \in R^{2 \times 2}.$$

Each term $X_i^\top \hat{r}_i$ is the score contribution from patient i — how much patient i is pulling $\hat{\beta}$ away from the truth. The meat sums the outer products of these contributions across all patients. If the model were correctly specified with independent observations, $\mathcal{M} \approx \mathcal{I}$ and the sandwich reduces to $\mathcal{B}$.

3.3 The Sandwich

Combining bread and meat gives the *sandwich* variance estimator :

$$\hat{V}_{\text{sand}} = \mathcal{B} \mathcal{M} \mathcal{B}.$$

This is the CR0 estimator . It is consistent as $m \rightarrow \infty$ but can be negatively biased when m is small, as we discuss in the next section.

4 The CR2 Adjustment

4.1 Why the Plain Sandwich is Biased

The sandwich estimator $\hat{V}_{\text{sand}}$ relies on the empirical outer products $\hat{r}_i \hat{r}_i^\top$ to estimate the true residual covariance $\text{Cov}(Y_i)$. However, fitted residuals are systematically shrunk by leverage. To see this, note that to first order:

$$\hat{r}_i \doteq (I - H_X)_i r,$$

where $r = (r_1^\top, \dots, r_m^\top)^\top$ are the stacked population residuals, $r_i = Y_i - \mu_i(\beta)$, and $(I - H_X)_i$ denotes the $N_i \times N$ submatrix of rows of $(I - H_X)$ belonging to patient i . The subscript i does *not* denote the $N_i \times N_i$ diagonal block — the full N -dimensional vector r appears because fitted residuals for patient i depend on population residuals from all patients through the off-diagonal blocks of $(I - H_X)$.

Taking the outer product and expectation, and using independence across patients so that $E[r_j r_k^\top] = 0$ for $j \neq k$ and $E[r_i r_i^\top] = V_i$, gives:

$$\begin{aligned} Q_i &:= E[\hat{r}_i \hat{r}_i^\top] \doteq (I - H_X)_i E[r r^\top] (I - H_X)_i^\top \\ &= (I - H_X)_i \mathbf{V} (I - H_X)_i^\top \\ &= \sum_{j=1}^m (I - H_X)_{ij} V_j (I - H_X)_{ij}^\top, \end{aligned} \tag{1}$$

where $(I - H_X)_{ij} = \delta_{ij} I - V_i X_i \mathcal{B} X_j^\top$ denotes the (i, j) block of $(I - H_X)$ of dimension $N_i \times N_j$.

Remark 1 *The cross-patient terms $j \neq i$ in (1) do not vanish in general, even though patients are independent. Independence kills $E[r_j r_k^\top]$ for $j \neq k$, but the off-diagonal blocks $(I - H_X)_{ij}$ for $j \neq i$ are non-zero constants determined by the design. These terms vanish only under the special case $\mathbf{V} = \sigma^2 I$ (Bell and McCaffrey, 2002), where the sum collapses to $(I - H_X)_{ii} \sigma^2 (I - H_X)_{ii}^\top$. For a general working variance $\mathbf{V}$, all m terms contribute.*

The fitted residuals $\hat{r}_i$ are therefore systematically shrunk relative to the true errors r_i , leading to:

$$E[\hat{V}_{\text{sand}}] \preceq \text{Var}(\hat{\beta}),$$

meaning the plain sandwich systematically underestimates the true variance. This bias is exacerbated when m is small or leverage is unevenly distributed across patients.

4.2 The BRL Criterion

Bias-Reduced Linearisation (BRL) seeks a cluster-specific matrix A_i such that replacing $\hat{r}_i$ with $A_i \hat{r}_i$ in the meat restores exact unbiasedness under the working model. The requirement is that the adjusted meat contribution from patient i has the correct expectation:

$$E[X_i^\top A_i \hat{r}_i \hat{r}_i^\top A_i^\top X_i] = X_i^\top V_i X_i.$$

Substituting $E[\hat{r}_i \hat{r}_i^\top] = Q_i$, this holds if and only if A_i satisfies:

$$A_i Q_i A_i^\top = V_i, \tag{2}$$

where Q_i is as defined in (1). We seek the unique symmetric positive definite solution to (2).

4.3 Solution via the Inner Matrix G_i

Define the inner matrix:

$$G_i = D_i Q_i D_i = D_i (I - H_X)_i \mathbf{V} (I - H_X)_i^\top D_i, \quad (3)$$

where $D_i = V_i^{1/2} = \text{diag}(\sqrt{\mu_i})$ is the symmetric square root of V_i . Since $Q_i \succ 0$ (under the mild condition that $(I - H_X)_i$ has full row rank, which holds when $N_i \leq N - p$) and $D_i \succ 0$, the matrix G_i is symmetric positive definite and $G_i^{-1/2}$ exists.

Proposition 1 *The matrix*

$$A_i = D_i G_i^{-1/2} D_i \quad (4)$$

is the unique symmetric positive definite solution to (2).

Substituting (4) into the left side of (2):

$$A_i Q_i A_i^\top = (D_i G_i^{-1/2} D_i) Q_i (D_i G_i^{-1/2} D_i)^\top.$$

Since D_i is symmetric diagonal, $D_i^\top = D_i$, so:

$$= D_i G_i^{-1/2} \underbrace{(D_i Q_i D_i)}_{=G_i} G_i^{-1/2} D_i = D_i G_i^{-1/2} G_i G_i^{-1/2} D_i = D_i^2 = V_i.$$

Symmetry of A_i follows because $G_i^{-1/2}$ is symmetric and D_i is symmetric diagonal, so $D_i G_i^{-1/2} D_i$ is symmetric. Positive definiteness follows because $D_i \succ 0$ and $G_i^{-1/2} \succ 0$.

Uniqueness: if $\tilde{A}_i$ is another symmetric PD solution, then setting $\tilde{C}_i = D_i^{-1} \tilde{A}_i D_i^{-1}$ yields $\tilde{C}_i G_i \tilde{C}_i = I$, so $\tilde{C}_i = G_i^{-1/2}$ by uniqueness of the symmetric square root, giving $\tilde{A}_i = A_i$.

Remark 2 (Whitening interpretation) *The solution (4) admits an equivalent derivation via a change of variables. Define:*

$$C_i = V_i^{-1/2} A_i V_i^{-1/2}, \quad F_i = V_i^{1/2} Q_i V_i^{1/2}.$$

Premultiplying and postmultiplying (2) by $V_i^{-1/2}$ transforms the criterion to $C_i F_i C_i^\top = I$, solved by $C_i = F_i^{-1/2}$, giving $A_i = V_i^{1/2} F_i^{-1/2} V_i^{1/2}$. Since $F_i = V_i^{1/2} Q_i V_i^{1/2} = D_i Q_i D_i = G_i$, this recovers exactly (4). The whitening construction is thus an alternative route to the same solution.

4.4 The CR2 Sandwich Estimator

Replacing $\hat{r}_i$ with $A_i \hat{r}_i$ in the meat gives the CR2-adjusted meat :

$$\widetilde{\mathcal{M}} = \sum_{i=1}^m X_i^\top A_i \hat{r}_i \hat{r}_i^\top A_i^\top X_i \in R^{2 \times 2}.$$

The CR2 sandwich estimator is:

$$\hat{V}_{\text{CR2}} = \mathcal{B} \widetilde{\mathcal{M}} \mathcal{B}.$$

By construction, $E[\hat{V}_{\text{CR2}}] = \text{Var}(\hat{\beta})$ under the working model $\text{Cov}(Y_i) = V_i$, so the estimator is exactly unbiased. It remains consistent as $m \rightarrow \infty$ under arbitrary true covariance structures, providing reliable inference even when the number of patients is small.

4.5 Simplification of G_i for the Poisson GLM

The full cross-cluster sum in Q_i collapses to a single diagonal-block expression for the Poisson GLM, due to the structure of the Fisher information matrix $\mathcal{I} = \sum_j X_j^\top V_j X_j$.

Proposition 2 *For the Poisson GLM with log link,*

$$Q_i = (I - H_{X,ii}) V_i, \quad (5)$$

and therefore:

$$G_i = D_i (I - H_{X,ii}) V_i D_i. \quad (6)$$

Substituting $(I - H_X)_{ij} = \delta_{ij}I - V_i X_i \mathcal{B} X_j^\top$ into the definition of Q_i :

$$Q_i = \sum_{j=1}^m (\delta_{ij}I - V_i X_i \mathcal{B} X_j^\top) V_j (\delta_{ij}I - X_j \mathcal{B} X_i^\top V_i).$$

Expanding the product gives four terms for each j . The three terms carrying a factor of δ_{ij} survive only at $j = i$; the remaining term sums freely over all j :

$$Q_i = V_i - V_i X_i \mathcal{B} X_i^\top V_i - V_i X_i \mathcal{B} X_i^\top V_i + V_i X_i \mathcal{B} \left(\sum_{j=1}^m X_j^\top V_j X_j \right) \mathcal{B} X_i^\top V_i.$$

Recognising $\sum_j X_j^\top V_j X_j = \mathcal{I}$ and using $\mathcal{B}\mathcal{I} = I$:

$$Q_i = V_i - 2V_i X_i \mathcal{B} X_i^\top V_i + V_i X_i \mathcal{B} X_i^\top V_i = V_i - V_i X_i \mathcal{B} X_i^\top V_i = (I - V_i X_i \mathcal{B} X_i^\top) V_i = (I - H_{X,ii}) V_i.$$

Therefore $G_i = D_i Q_i D_i = D_i (I - H_{X,ii}) V_i D_i$.

5 Closed-Form $G_i^{-1/2}$ Under One Image Per Patient

The standard CR2 computation requires an eigen-decomposition of G_i at $O(N_i^3)$ cost per patient. When each patient contributes exactly one image ($n_i = 1$ for all i), the spicyGLM design forces G_i to be compound symmetric, admitting a closed-form $O(1)$ expression for $G_i^{-1/2}$ and eliminating the eigen-decomposition entirely.

5.1 Matrix Structure Under One Image Per Patient

When $n_i = 1$, all N_i REF cells in patient i share the same image-level offset $\log(\widehat{\text{den}}_{i1})$ and the same group indicator $g(i)$. Every cell therefore has the same fitted mean:

$$\hat{\mu}_{i1k} = \exp(\hat{\beta}_{g(i)}) \times \widehat{\text{den}}_{i1} =: \hat{\mu}_i \quad \text{for all } k = 1, \dots, N_i.$$

The per-patient matrices therefore simplify to:

$$V_i = \hat{\mu}_i I_{N_i}, \quad D_i = \hat{\mu}_i^{1/2} I_{N_i},$$

and the design matrix has rank 1 with all rows identical:

$$X_i = \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top.$$

5.2 Information Matrix and Bread

For patient i , the contribution to the Fisher information matrix is:

$$X_i^\top V_i X_i = \mathbf{e}_{g(i)} \mathbf{1}_{N_i}^\top \cdot \hat{\mu}_i I_{N_i} \cdot \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top = N_i \hat{\mu}_i \mathbf{e}_{g(i)} \mathbf{e}_{g(i)}^\top.$$

Since $\mathbf{e}_{g(i)} \mathbf{e}_{g(i)}^\top$ equals either $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ or $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ depending on group membership, the cross terms vanish when summing over patients and the sum separates by group:

$$\mathcal{I} = \sum_{i=1}^m N_i \hat{\mu}_i \mathbf{e}_{g(i)} \mathbf{e}_{g(i)}^\top = \sum_{i: g(i)=1} N_i \hat{\mu}_i \mathbf{e}_1 \mathbf{e}_1^\top + \sum_{i: g(i)=2} N_i \hat{\mu}_i \mathbf{e}_2 \mathbf{e}_2^\top = \begin{pmatrix} S_1 & 0 \\ 0 & S_2 \end{pmatrix},$$

where $S_g = \sum_{i: g(i)=g} N_i \hat{\mu}_i$. Therefore:

$$\mathcal{B} = \mathcal{I}^{-1} = \begin{pmatrix} 1/S_1 & 0 \\ 0 & 1/S_2 \end{pmatrix}.$$

5.3 Leverage Scalar α_i

The diagonal block of the hat matrix is:

$$H_{X,ii} = V_i X_i \mathcal{B} X_i^\top = \hat{\mu}_i \mathbf{1}_{N_i} \underbrace{\mathbf{e}_{g(i)}^\top \mathcal{B} \mathbf{e}_{g(i)}}_{=1/S_{g(i)}} \mathbf{1}_{N_i}^\top = \alpha_i \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top,$$

where:

$$\alpha_i = \frac{\hat{\mu}_i}{S_{g(i)}}.$$

Therefore:

$$I_{N_i} - H_{X,ii} = I_{N_i} - \alpha_i \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top,$$

which is compound symmetric with diagonal entries $1 - \alpha_i$ and off-diagonal entries $-\alpha_i$.

Proposition 3 $0 < \alpha_i < 1/N_i$ for all patients i , provided patient i 's group contains at least one other patient.

Since $\hat{\mu}_i > 0$ and $S_{g(i)} > 0$, clearly $\alpha_i > 0$. For the upper bound, $S_{g(i)} = \sum_{j: g(j)=g(i)} N_j \hat{\mu}_j \geq N_i \hat{\mu}_i$, so $\alpha_i = \hat{\mu}_i / S_{g(i)} \leq 1/N_i$, with equality if and only if patient i is the sole member of its group (i.e. $S_{g(i)} = N_i \hat{\mu}_i$ exactly, with no other patient's contribution to the sum). Whenever the group contains at least one further patient $j \neq i$, $S_{g(i)} > N_i \hat{\mu}_i$ strictly, since $N_j \hat{\mu}_j > 0$, giving strict inequality.

Remark 3 The bound $\alpha_i < 1/N_i$ (Proposition 3) ensures $1 - \alpha_i N_i > 0$, which guarantees $G_i \succ 0$ and $G_i^{-1/2}$ exists. Consequently, CR2 is only well-defined for groups containing at least two patients: if patient i is the sole member of group $g(i)$, then $\alpha_i N_i = 1$ exactly, G_i becomes singular, and the adjustment matrix A_i does not exist.

5.4 Structure of G_i

Applying the simplification from Proposition 2 and substituting $D_i = \hat{\mu}_i^{1/2}I$, $V_i = \hat{\mu}_i I$, and $H_{X,ii} = \alpha_i \mathbf{1}\mathbf{1}^\top$:

$$\begin{aligned} G_i &= D_i (I - H_{X,ii}) V_i D_i \\ &= \hat{\mu}_i^{1/2} I \cdot (I - \alpha_i \mathbf{1}\mathbf{1}^\top) \cdot \hat{\mu}_i I \cdot \hat{\mu}_i^{1/2} I \\ &= \hat{\mu}_i^2 (I - \alpha_i \mathbf{1}\mathbf{1}^\top). \end{aligned}$$

G_i is compound symmetric with diagonal entries $\hat{\mu}_i^2(1 - \alpha_i)$ and off-diagonal entries $-\hat{\mu}_i^2\alpha_i$.

5.5 Eigenvalues of G_i

Lemma 1 $G_i = \hat{\mu}_i^2(I - \alpha_i \mathbf{1}\mathbf{1}^\top)$ has exactly two distinct eigenvalues:

1. $\lambda_1 = \hat{\mu}_i^2(1 - \alpha_i N_i)$, with eigenvector $\mathbf{1}_{N_i}/\sqrt{N_i}$ (multiplicity 1),
2. $\lambda_2 = \hat{\mu}_i^2$, with eigenvectors $v \perp \mathbf{1}_{N_i}$ (multiplicity $N_i - 1$).

Both eigenvalues are strictly positive by Proposition 3.

For λ_1 :

$$G_i \mathbf{1} = \hat{\mu}_i^2(I - \alpha_i \mathbf{1}\mathbf{1}^\top) \mathbf{1} = \hat{\mu}_i^2(\mathbf{1} - \alpha_i N_i \mathbf{1}) = \hat{\mu}_i^2(1 - \alpha_i N_i) \mathbf{1}.$$

For λ_2 : for any v with $\mathbf{1}^\top v = 0$,

$$G_i v = \hat{\mu}_i^2 \left(v - \alpha_i \mathbf{1} \underbrace{\mathbf{1}^\top v}_{=0} \right) = \hat{\mu}_i^2 v.$$

5.6 Closed-Form $G_i^{-1/2}$

By the spectral theorem, decomposing into the $\mathbf{1}$ -direction and its orthogonal complement:

$$G_i^{-1/2} = \lambda_1^{-1/2} \frac{\mathbf{1}\mathbf{1}^\top}{N_i} + \lambda_2^{-1/2} \left(I - \frac{\mathbf{1}\mathbf{1}^\top}{N_i} \right).$$

Substituting $\lambda_1^{-1/2} = \hat{\mu}_i^{-1}(1 - \alpha_i N_i)^{-1/2}$ and $\lambda_2^{-1/2} = \hat{\mu}_i^{-1}$ (Lemma 1):

$$\begin{aligned} G_i^{-1/2} &= \frac{1}{\hat{\mu}_i \sqrt{1 - \alpha_i N_i}} \frac{\mathbf{1}\mathbf{1}^\top}{N_i} + \frac{1}{\hat{\mu}_i} \left(I - \frac{\mathbf{1}\mathbf{1}^\top}{N_i} \right) \\ &= \frac{1}{\hat{\mu}_i} I + \frac{1}{\hat{\mu}_i N_i} \left(\frac{1}{\sqrt{1 - \alpha_i N_i}} - 1 \right) \mathbf{1}\mathbf{1}^\top. \end{aligned} \tag{7}$$

5.7 The CR2 Adjustment Matrix A_i

Applying $A_i = D_i G_i^{-1/2} D_i$ with $D_i = \hat{\mu}_i^{1/2} I$:

$$A_i = \hat{\mu}_i^{1/2} I \cdot G_i^{-1/2} \cdot \hat{\mu}_i^{1/2} I = \hat{\mu}_i G_i^{-1/2}.$$

Substituting (7) and noting that $\hat{\mu}_i$ cancels completely:

$$A_i = I_{N_i} + \frac{1}{N_i} \left(\frac{1}{\sqrt{1 - \alpha_i N_i}} - 1 \right) \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top, \quad (8)$$

where $\alpha_i = \hat{\mu}_i / S_{g(i)}$ and $S_g = \sum_{i: g(i)=g} N_i \hat{\mu}_i$. The matrix A_i is compound symmetric and characterised entirely by the scalar α_i and the cluster size N_i ; no $N_i \times N_i$ matrix needs to be formed or eigen-decomposed.

5.8 Computational Algorithm

Algorithm 1 Closed-Form CR2 Adjustment (One Image Per Patient)

Require: Fitted spicyGLM, fitted means $\{\hat{\mu}_i\}$, cell counts $\{N_i\}$, group membership $\{g(i)\}$, residual vectors $\{\hat{r}_i\}$

Ensure: $A_i \hat{r}_i$ for each patient i , via $A_i = I_{N_i} + \frac{1}{N_i} ((1 - \alpha_i N_i)^{-1/2} - 1) \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top$

- 1: $S_g \leftarrow \sum_{i: g(i)=g} N_i \hat{\mu}_i$ for $g \in \{1, 2\}$ $\triangleright O(m)$
- 2: **for** each patient $i = 1, \dots, m$ **do**
- 3: $\alpha_i \leftarrow \hat{\mu}_i / S_{g(i)}$ $\triangleright$ leverage scalar, $O(1)$
- 4: $c_i \leftarrow \frac{1}{N_i} ((1 - \alpha_i N_i)^{-1/2} - 1)$ $\triangleright O(1)$
- 5: $s_i \leftarrow \mathbf{1}_{N_i}^\top \hat{r}_i$ $\triangleright$ sum of residuals, $O(N_i)$
- 6: $A_i \hat{r}_i \leftarrow \hat{r}_i + c_i s_i \mathbf{1}_{N_i}$ $\triangleright O(N_i)$
- 7: **end for**

6 Closed-Form $G_i^{-1/2}$ Under Multiple Images Per Patient

6.1 Matrix Structure Under Multiple Images Per Patient

When patient i has $n_i \geq 1$ images, cells within the same image j share the same fitted mean, but this mean generally varies across images. Specifically, for cell k in image j of patient i :

$$\hat{\mu}_{ijk} = \exp(\hat{\beta}_{g(i)}) \times \widehat{\text{den}}_{ij} =: \hat{\mu}_{ij} \quad \text{for all } k = 1, \dots, n_{ij}.$$

Since group membership $g(i)$ is patient-level, all images of patient i share the same $\hat{\beta}_{g(i)}$; variation in $\hat{\mu}_{ij}$ across images j arises solely from the image-level density offset $\widehat{\text{den}}_{ij}$.

The per-patient working variance matrix is therefore block-diagonal, constant within each image block but varying across blocks:

$$V_i = \text{blkdiag} \left(\hat{\mu}_{i1} I_{n_{i1}}, \dots, \hat{\mu}_{i,n_i} I_{n_{i,n_i}} \right),$$

with corresponding $D_i = V_i^{1/2} = \text{blkdiag}(\hat{\mu}_{i1}^{1/2} I_{n_{i1}}, \dots)$. We write $\boldsymbol{\mu}_i \in R^{N_i}$ for the vector of fitted means (constant on each image block), so $V_i = \text{diag}(\boldsymbol{\mu}_i)$ as in the general notation of Table 2.

The design matrix retains its rank-1 structure regardless of n_i , since group membership does not vary within a patient:

$$X_i = \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top.$$

This is the key structural fact enabling the closed-form results below: although V_i is no longer a scalar multiple of the identity (as it was for $n_i = 1$ in Section 5), it remains constant on a coarser partition of the N_i cells — the n_i images — and X_i is unaffected by the number of images entirely.

6.2 The Image-Indicator Subspace and Its Complement

The closed-form results in this section rely on a specific orthonormal decomposition of R^{N_i} , adapted to the block structure of V_i established in Section 6.1. We construct this decomposition here before deriving the structure of G_i itself.

For each image $j = 1, \dots, n_i$, define the normalized indicator vector $p_{ij} \in R^{N_i}$, equal to $1/\sqrt{n_{ij}}$ on the n_{ij} coordinates belonging to image j and zero elsewhere. Collect these as columns of

$$P_i = [p_{i1} \ p_{i2} \ \dots \ p_{i,n_i}] \in R^{N_i \times n_i}.$$

Since distinct images occupy disjoint blocks of coordinates, the columns of P_i are orthonormal:

$$P_i^\top P_i = I_{n_i}.$$

Note that $P_i P_i^\top \neq I_{N_i}$ in general (equality holds only if $n_{ij} = 1$ for all j): $P_i P_i^\top$ is instead the orthogonal projector onto the n_i -dimensional subspace $\text{range}(P_i) \subset R^{N_i}$, which we call the *image-indicator subspace* — the space of vectors that are constant within each image.

Let $R_i \in R^{N_i \times (N_i - n_i)}$ denote any matrix whose columns form an orthonormal basis for the orthogonal complement of $\text{range}(P_i)$ in R^{N_i} . Concretely, this complement decomposes image-by-image: for each image j , the sum-to-zero subspace $\{v \in R^{n_{ij}} : \mathbf{1}_{n_{ij}}^\top v = 0\}$, embedded in R^{N_i} by zero-padding outside image j 's block, contributes $n_{ij} - 1$ dimensions; summing over images gives $\sum_j (n_{ij} - 1) = N_i - n_i$ dimensions in total, matching the size of R_i . Any orthonormal basis of each image's sum-to-zero subspace may be used (e.g. Helmert contrasts); the specific choice is immaterial to what follows; see Proposition 4 below.

By construction, $R_i^\top P_i = 0$, and

$$U_i := [P_i \ R_i] \in R^{N_i \times N_i}$$

is an orthogonal matrix, i.e. $U_i^\top U_i = U_i U_i^\top = I_{N_i}$, with the complementary projector identity

$$R_i R_i^\top = I_{N_i} - P_i P_i^\top.$$

When $n_i = 1$, P_i reduces to the single column $\mathbf{1}_{N_i}/\sqrt{N_i}$ used throughout Section 5, and R_i reduces to the $(N_i - 1)$ -dimensional complement used in Lemma 1.

6.3 Structure of G_i : The DPR1 Form

We now derive the structure of G_i under multiple images per patient, generalizing Sections 5.2–5.4. Throughout, we use the Poisson-GLM simplification $Q_i = (I - H_{X,ii})V_i$ established in Proposition 2.

Information matrix and bread. The contribution of patient i to the Fisher information matrix is

$$X_i^\top V_i X_i = \mathbf{e}_{g(i)}^\top \mathbf{1}_{N_i}^\top V_i \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top = T_i \mathbf{e}_{g(i)} \mathbf{e}_{g(i)}^\top, \quad T_i := \mathbf{1}_{N_i}^\top V_i \mathbf{1}_{N_i} = \sum_{j=1}^{n_i} n_{ij} \hat{\mu}_{ij}.$$

As in Section 5.2, the cross terms vanish when summing over patients, giving $\mathcal{I} = \text{diag}(S_1, S_2)$ with $S_g = \sum_{i:g(i)=g} T_i$, and $\mathcal{B} = \text{diag}(1/S_1, 1/S_2)$.

The hat matrix block $H_{X,ii}$. Using $\mathcal{B}$ diagonal and $\mathbf{e}_{g(i)}^\top \mathbf{e}_{g(i)} = 1$:

$$H_{X,ii} = V_i X_i \mathcal{B} X_i^\top = V_i \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top \mathcal{B} \mathbf{e}_{g(i)} \mathbf{1}_{N_i}^\top = \frac{1}{S_{g(i)}} V_i \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top = \frac{1}{S_{g(i)}} \boldsymbol{\mu}_i \mathbf{1}_{N_i}^\top,$$

using $V_i \mathbf{1}_{N_i} = \boldsymbol{\mu}_i$. Note $H_{X,ii}$ is not symmetric in general (it is an outer product of $\boldsymbol{\mu}_i$ with $\mathbf{1}_{N_i}$, not with itself); this causes no difficulty, as H_X need not be symmetric — only Q_i , derived next, is required to be.

Q_i and G_i . By Proposition 2, $Q_i = (I - H_{X,ii})V_i = V_i - H_{X,ii}V_i$. Since V_i is diagonal, $\mathbf{1}_{N_i}^\top V_i = \boldsymbol{\mu}_i^\top$, so:

$$Q_i = V_i - \frac{1}{S_{g(i)}} \boldsymbol{\mu}_i (\mathbf{1}_{N_i}^\top V_i) = V_i - \frac{1}{S_{g(i)}} \boldsymbol{\mu}_i \boldsymbol{\mu}_i^\top,$$

which is symmetric, as required. Applying $D_i = \text{diag}(\boldsymbol{\mu}_i^{1/2})$ on both sides:

$$G_i = D_i Q_i D_i = D_i V_i D_i - \frac{1}{S_{g(i)}} (D_i \boldsymbol{\mu}_i) (D_i \boldsymbol{\mu}_i)^\top.$$

Since $D_i V_i D_i = \text{diag}(\boldsymbol{\mu}_i) \cdot \text{diag}(\boldsymbol{\mu}_i) = \text{diag}(\boldsymbol{\mu}_i^2) =: \Lambda_i$, and $D_i \boldsymbol{\mu}_i = \text{diag}(\boldsymbol{\mu}_i^{1/2}) \boldsymbol{\mu}_i = \boldsymbol{\mu}_i^{3/2} =: \mathbf{f}_i$ (all operations elementwise), we obtain:

$$G_i = \Lambda_i - \frac{1}{S_{g(i)}} \mathbf{f}_i \mathbf{f}_i^\top, \quad \Lambda_i = \text{diag}(\boldsymbol{\mu}_i^2), \quad \mathbf{f}_i = \boldsymbol{\mu}_i^{3/2}. \quad (9)$$

This is a diagonal-plus-rank-one (DPR1) matrix. Crucially, both Λ_i and $\mathbf{f}_i$ are *constant on each image block*: entry $\hat{\mu}_{ij}^2$ (resp. $\hat{\mu}_{ij}^{3/2}$) is repeated across all n_{ij} cells of image j . Consequently Λ_i has only n_i distinct diagonal values — a repeated-pole structure that is exploited throughout the remainder of this section. When $n_i = 1$, (9) reduces to $G_i = \hat{\mu}_i^2 I - \frac{\hat{\mu}_i^3}{S_{g(i)}} \mathbf{1}\mathbf{1}^\top = \hat{\mu}_i^2 (I - \alpha_i N_i \cdot \frac{1}{N_i} \mathbf{1}\mathbf{1}^\top)$, recovering the compound-symmetric form of Section 5.4 (using $S_{g(i)} = N_i \hat{\mu}_i / \alpha_i$ from the definition of α_i , hence $\hat{\mu}_i^3 / S_{g(i)} = \hat{\mu}_i^2 \alpha_i$).

6.4 Closed-Form Eigenpairs from the Within-Image Complement

We first identify $N_i - n_i$ eigenpairs of G_i in closed form, generalizing Lemma 1.

Proposition 4 *Let $r \in R^{N_i}$ be any column of R_i (Section 6.2): r is supported on a single image j 's block of coordinates, and sums to zero within that block. Then r is an eigenvector of G_i with eigenvalue $\hat{\mu}_{ij}^2$.*

Write $G_i r = \Lambda_i r - \frac{1}{S_{g(i)}} \mathbf{f}_i (\mathbf{f}_i^\top r)$, and evaluate each term.

Since Λ_i is diagonal with value $\hat{\mu}_{ij}^2$ on every coordinate of image j 's block, and r vanishes outside that block,

$$\Lambda_i r = \hat{\mu}_{ij}^2 r.$$

For the second term, $\mathbf{f}_i$ is likewise constant ($= \hat{\mu}_{ij}^{3/2}$) on image j 's block and r vanishes elsewhere, so

$$\mathbf{f}_i^\top r = \hat{\mu}_{ij}^{3/2} \sum_{\text{cells in image } j} r_{\text{cell}} = \hat{\mu}_{ij}^{3/2} \cdot 0 = 0,$$

using the defining property of r (it sums to zero within image j). Hence

$$G_i r = \hat{\mu}_{ij}^2 r - \frac{1}{S_{g(i)}} \mathbf{f}_i \cdot 0 = \hat{\mu}_{ij}^2 r.$$

Since this holds for every column of R_i , and any orthonormal basis of each image's sum-to-zero subspace may be used (Section 6.2), the entire $(n_{ij} - 1)$ -dimensional sum-to-zero subspace of image j is an eigenspace of G_i with eigenvalue $\hat{\mu}_{ij}^2$. Summing over images, this accounts for

$$\sum_{j=1}^{n_i} (n_{ij} - 1) = N_i - n_i$$

eigenpairs of G_i .

When $n_i = 1$, image 1's sum-to-zero subspace is all of $\{v : \mathbf{1}_{N_i}^\top v = 0\}$, and Proposition 4 reduces exactly to the $\lambda_2 = \hat{\mu}_i^2$ eigenspace of Lemma 1.

6.5 The Reduced Eigenvalue Problem on the Image-Indicator Subspace

Proposition 4 accounts for $N_i - n_i$ of the N_i eigenpairs of G_i ; the remaining n_i eigenpairs are shown here to coincide with the eigenpairs of a small $n_i \times n_i$ matrix $\bar{G}_i$, via the image-indicator subspace $\text{range}(P_i)$ introduced in Section 6.2.

Proposition 5 (Invariance) *$\text{range}(P_i)$ is invariant under G_i : for every $j = 1, \dots, n_i$, $G_i p_{ij} \in \text{range}(P_i)$.*

Write $G_i p_{ij} = \Lambda_i p_{ij} - \frac{1}{S_{g(i)}} \mathbf{f}_i (\mathbf{f}_i^\top p_{ij})$. Since Λ_i is constant ($= \hat{\mu}_{ij}^2$) on image j 's block and p_{ij} is supported there, $\Lambda_i p_{ij} = \hat{\mu}_{ij}^2 p_{ij} \in \text{range}(P_i)$. For the second term, $\mathbf{f}_i^\top p_{ij}$ is a scalar, and $\mathbf{f}_i$ itself lies in $\text{range}(P_i)$: since $\mathbf{f}_i$ is constant on every image block, $\mathbf{f}_i = \sum_{k=1}^{n_i} \sqrt{n_{ik}} \hat{\mu}_{ik}^{3/2} p_{ik}$, a linear combination of P_i 's columns. Hence $\mathbf{f}_i (\mathbf{f}_i^\top p_{ij}) \in \text{range}(P_i)$ as well, and $G_i p_{ij} \in \text{range}(P_i)$.

Definition 1 *The reduced inner matrix is*

$$\bar{G}_i := P_i^\top G_i P_i \in R^{n_i \times n_i}.$$

Proposition 6 (Reduced DPR1 form)

$$\bar{G}_i = \bar{\Lambda}_i - \frac{1}{S_{g(i)}} \bar{\mathbf{f}}_i \bar{\mathbf{f}}_i^\top, \quad \bar{\Lambda}_i = \text{diag}(\hat{\mu}_{i1}^2, \dots, \hat{\mu}_{i,n_i}^2), \quad (\bar{\mathbf{f}}_i)_j = \sqrt{n_{ij}} \hat{\mu}_{ij}^{3/2}.$$

Entry (j, k) of $P_i^\top \Lambda_i P_i$ is $p_{ij}^\top \Lambda_i p_{ik}$. For $j \neq k$, p_{ij} and p_{ik} have disjoint support, so this vanishes; for $j = k$, $p_{ij}^\top \Lambda_i p_{ij} = \sum_{\text{cells in image } j} \frac{1}{n_{ij}} \hat{\mu}_{ij}^2 = \hat{\mu}_{ij}^2$. Hence $P_i^\top \Lambda_i P_i = \bar{\Lambda}_i$. Entry j of $P_i^\top \mathbf{f}_i$ is $p_{ij}^\top \mathbf{f}_i = \sum_{\text{cells in image } j} \frac{1}{\sqrt{n_{ij}}} \hat{\mu}_{ij}^{3/2} = \sqrt{n_{ij}} \hat{\mu}_{ij}^{3/2}$, giving $P_i^\top \mathbf{f}_i = \bar{\mathbf{f}}_i$. The claim follows from $\bar{G}_i = P_i^\top \Lambda_i P_i - \frac{1}{S_{g(i)}} (P_i^\top \mathbf{f}_i)(\mathbf{f}_i^\top P_i)$.

Lemma 2 (Lifting identity) $G_i P_i = P_i \bar{G}_i$.

By Proposition 5, $G_i P_i = P_i C$ for some $C \in R^{n_i \times n_i}$. Left-multiplying by $P_i^\top$ and using $P_i^\top P_i = I_{n_i}$ gives $C = P_i^\top G_i P_i = \bar{G}_i$.

Proposition 7 (Eigenpair lifting) *If $\bar{v} \in R^{n_i}$ satisfies $\bar{G}_i \bar{v} = \lambda \bar{v}$, then $P_i \bar{v}$ satisfies $G_i(P_i \bar{v}) = \lambda(P_i \bar{v})$.*

$G_i(P_i \bar{v}) = (G_i P_i) \bar{v} = (P_i \bar{G}_i) \bar{v} = P_i(\bar{G}_i \bar{v}) = P_i(\lambda \bar{v}) = \lambda(P_i \bar{v})$, using Lemma 2.

By Proposition 5, the n_i eigenpairs of G_i not already identified in Section 6.4 lie entirely within $\text{range}(P_i)$; by Proposition 7, they correspond exactly, via lifting through P_i , to the n_i eigenpairs of the small matrix $\bar{G}_i$. Note that $\bar{G}_i$ has the same DPR1 structure as G_i itself (Proposition 6), but of size $n_i \times n_i$ rather than $N_i \times N_i$ — solving for its eigenpairs is the subject of Section 6.6.

When $n_i = 1$, $\bar{G}_i$ is the 1×1 matrix $\hat{\mu}_{i1}^2(1 - \alpha_i N_i)$, recovering λ_1 of Lemma 1 directly (with no eigendecomposition required, as noted in Section 6.6 below).

6.6 Solving the Reduced Problem via DPR1 Theory

By Proposition 6, $\bar{G}_i$ is a diagonal-plus-rank-one (DPR1) matrix, and its eigendecomposition can be obtained via the theory of Jakovčević Stor et al. (2015) rather than a generic $O(n_i^3)$ eigensolver. This section states the conditions under which their theory applies and how it is invoked.

Matching to standard DPR1 form. Jakovčević Stor et al. (2015) study matrices $A = D + \rho z z^\top$ with the normalizing assumption $\rho > 0$. Matching term by term, $D \leftrightarrow \bar{\Lambda}_i$, $z \leftrightarrow \bar{\mathbf{f}}_i$, $\rho \leftrightarrow -1/S_{g(i)}$. Since $S_{g(i)} > 0$, we have $\rho < 0$, so we apply their stated remedy for this case: decompose $-\bar{G}_i = (-\bar{\Lambda}_i) + \frac{1}{S_{g(i)}} \bar{\mathbf{f}}_i \bar{\mathbf{f}}_i^\top$ instead, which has $\rho' = 1/S_{g(i)} > 0$. If $-\bar{G}_i = \bar{V}_i \bar{\Lambda}_i^{\text{eig}} \bar{V}_i^\top$ is the resulting eigendecomposition, then $\bar{G}_i = \bar{V}_i (-\bar{\Lambda}_i^{\text{eig}}) \bar{V}_i^\top$: eigenvectors are unchanged, eigenvalues are negated.

Regularity conditions. Their theory further requires: (i) all entries of z nonzero, and (ii) all diagonal entries of D distinct (*irreducibility*).

Condition (i) holds automatically here: $(\bar{\mathbf{f}}_i)_j = \sqrt{n_{ij}}\hat{\mu}_{ij}^{3/2} > 0$ for every image j , since $n_{ij} \geq 1$ and $\hat{\mu}_{ij} > 0$ always.

Condition (ii) requires $\hat{\mu}_{ij}^2 \neq \hat{\mu}_{ik}^2$ for all images $j \neq k$ of patient i — equivalently, since $\hat{\mu}_{ij} = \exp(\hat{\beta}_{g(i)})\widehat{\text{den}}_{ij}$ with $\hat{\beta}_{g(i)}$ common to all of patient i 's images, that no two images of the same patient share the same density offset $\widehat{\text{den}}_{ij}$. This is not guaranteed a priori and should be checked in practice, particularly for patients with many images.

Remark 4 (Tied poles) *If $m \geq 2$ images of patient i share a common value $\hat{\mu}_{ij}^2 = d$, the corresponding block admits $m - 1$ closed-form eigenpairs by the same mechanism as Proposition 4: any vector supported on the m tied coordinates and orthogonal to the corresponding entries of $\bar{\mathbf{f}}_i$ is an eigenvector with eigenvalue d . The remaining eigenvalue is obtained by collapsing the m tied coordinates to a single representative pole at d with effective entry $\tilde{\zeta} = (\sum \zeta_j^2)^{1/2}$ (sum over the tied coordinates), reducing the secular equation to size $n_i - (m - 1)$. This is the Givens-rotation-based reduction described in Jakovčević Stor et al. (2015, Section 1).*

A very short description of the `dpr1eig` algorithm. Given a DPR1 matrix in standard form with distinct poles, each eigenpair is obtained via

$$(\lambda_k, v_k) \leftarrow \text{dpr1eig}(D, z, \rho, k), \quad k = 1, \dots, n,$$

the algorithm of Jakovčević Stor et al. (2015).

Mechanism. Eigenvalues of a DPR1 matrix are roots of the secular equation

$$f(\lambda) = 1 + \rho \sum_{j=1}^n \frac{\zeta_j^2}{d_j - \lambda},$$

and satisfy a strict interlacing property with the poles, $\lambda_1 > d_1 > \lambda_2 > d_2 > \dots$, which brackets each root to a known interval between consecutive poles. Root-finding f directly within such a bracket (e.g. by bisection) is numerically fragile when poles are closely spaced: near a root, f is evaluated as a sum of terms that individually grow large and nearly cancel, so a rounding error incurred while forming those large terms can dominate the true (small) result. This is called a catastrophic cancellation.

`dpr1eig` avoids this by *shifting* to the pole d_i nearest the eigenvalue λ to be found. Rather than working with λ itself, it works directly with a small offset $\mu = \lambda - d_i$. Hence the goal now is to find the corresponding eigenvalue of the shifted matrix $A_i = A - d_i I$. Because μ is small by construction, it can be represented and manipulated directly, without first forming λ and d_i as two large, nearly equal quantities whose difference would otherwise be lost to cancellation. It then *inverts*: A_i^{-1} is a permuted arrowhead matrix with a known closed form, and $\nu = 1/\mu$ is precisely its largest-magnitude eigenvalue — a numerically well-conditioned quantity, obtained via an analogous secular equation on the arrowhead structure. The eigenvector follows from the standard DPR1 eigenvector formula applied to the shifted matrix. One scalar in the arrowhead inverse (the pivot entry b , eq. 8 of the

cited paper) retains its own, distinct cancellation risk when poles neighbouring the shift are closely spaced; the paper supplies explicit diagnostics (K_b, K_z, κ_ν) for when this scalar requires recomputation in extended precision, together with remedies for cases where the shifted pole itself is poorly chosen. We do not reproduce these diagnostics here and refer to Jakovčević Stor et al. (2015, Sections 2.1–2.3) for full details; they are incorporated directly into our implementation of **dpr1eig** without modification.

Applied to $\bar{G}_i$, **dpr1eig** runs in $O(n_i)$ operations per eigenpair, i.e. $O(n_i^2)$ total — a substantial reduction from the $O(n_i^3)$ cost of a generic eigensolver applied to $\bar{G}_i$, and negligible relative to the $O(N_i^3)$ cost of eigendecomposing G_i directly (Section 6.9).

Algorithm 2 Reduced eigendecomposition of $\bar{G}_i$ via **dpr1eig**

Require: Fitted means $\{\hat{\mu}_{ij}\}_{j=1}^{n_i}$, number of cells per image $\{n_{ij}\}_{j=1}^{n_i}$, group total $S_{g(i)} = \sum_{i':g(i')=g(i)} T_{i'}$ (Section 6.3)

Ensure: Eigenpairs $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i}$ of $\bar{G}_i = \bar{\Lambda}_i - \frac{1}{S_{g(i)}} \bar{\mathbf{f}}_i \bar{\mathbf{f}}_i^\top$

- 1: $\bar{\Lambda}_i \leftarrow \text{diag}(\hat{\mu}_{i1}^2, \dots, \hat{\mu}_{i,n_i}^2)$; $(\bar{\mathbf{f}}_i)_j \leftarrow \sqrt{n_{ij}} \hat{\mu}_{ij}^{3/2}$ ▷ Proposition 6, $O(n_i)$
 - 2: Sort images by $\hat{\mu}_{ij}^2$ descending; permute $\bar{\Lambda}_i, \bar{\mathbf{f}}_i$ accordingly; record permutation π ▷ ordering condition, below
 - 3: **if** any $\hat{\mu}_{ij}^2 = \hat{\mu}_{ik}^2$ (tied poles) **then**
 - 4: Apply the reduction of Remark 4: extract $m - 1$ closed-form eigenpairs per tied group, collapse each group to one representative pole
 - 5: **end if**
 - 6: $D \leftarrow -\bar{\Lambda}_i$; $z \leftarrow \bar{\mathbf{f}}_i$; $\rho \leftarrow 1/S_{g(i)}$ ▷ sign flip, above
 - 7: **for** $k = 1, \dots, n_i$ **do** ▷ (reduced size, after Step 3 if triggered)
 - 8: $(\lambda_k^-, v_k) \leftarrow \text{dpr1eig}(D, z, \rho, k)$ ▷ Algorithm 4
 - 9: $\bar{\lambda}_k \leftarrow -\lambda_k^-$; $\bar{v}_k \leftarrow v_k$ ▷ undo sign flip
 - 10: **end for**
 - 11: Un-permute $(\bar{\lambda}_k, \bar{v}_k)$ via π^{-1} ; re-insert any closed-form pairs from Step 3
 - 12: **return** $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i}$
-

Ordering. The theory further assumes poles are sorted in decreasing order; since eigenvalues are unaffected by a simultaneous relabelling of D 's diagonal and z 's entries, this is achieved by sorting the n_i images of patient i by $\hat{\mu}_{ij}^2$ before invoking **dpr1eig**, and un-sorting the resulting eigenvector indices when lifting via P_i (Section 6.7).

6.7 Closed-Form Assembly of $G_i^{-1/2}$

We now combine the eigenpairs from Sections 6.4 and 6.6 into a single closed-form expression for $G_i^{-1/2}$.

Proposition 8

$$G_i^{-1/2} = \Lambda_i^{-1/2} (I_{N_i} - P_i P_i^\top) + P_i \bar{V}_i \left(\bar{\Lambda}_i^{\text{eig}} \right)^{-1/2} \bar{V}_i^\top P_i^\top,$$

where $\bar{V}_i \in R^{n_i \times n_i}$ is orthogonal with columns the eigenvectors of $\bar{G}_i$, and $\bar{\Lambda}_i^{\text{eig}} = \text{diag}(\bar{\lambda}_1, \dots, \bar{\lambda}_{n_i})$ the corresponding eigenvalues, both obtained via **dpr1eig** (Section 6.6).

Let $U_i = [R_i \ P_i \bar{V}_i]$, an $N_i \times N_i$ orthogonal matrix: its first $N_i - n_i$ columns are the closed-form eigenvectors of Proposition 4, and its remaining n_i columns are the lifted eigenvectors of $\bar{G}_i$ (Proposition 7), which are mutually orthogonal since $\bar{V}_i$ is orthogonal and $P_i^\top P_i = I_{n_i}$, and orthogonal to R_i 's columns since $R_i^\top P_i = 0$. By the spectral theorem,

$$G_i^{-1/2} = U_i \begin{bmatrix} (\Lambda_i^R)^{-1/2} & 0 \\ 0 & (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \end{bmatrix} U_i^\top = R_i (\Lambda_i^R)^{-1/2} R_i^\top + P_i \bar{V}_i (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \bar{V}_i^\top P_i^\top,$$

where Λ_i^R holds the eigenvalues from Proposition 4 (block-diagonal, $\hat{\mu}_{ij}^2$ on the columns of R_i belonging to image j), and the cross terms vanish as the middle matrix is block-diagonal. It remains to show

$$R_i (\Lambda_i^R)^{-1/2} R_i^\top = \Lambda_i^{-1/2} (I_{N_i} - P_i P_i^\top). \quad (10)$$

Each column of R_i is supported entirely within a single image block, on which Λ_i acts as the scalar $\hat{\mu}_{ij}^2$; hence $\Lambda_i^{-1/2}$ scales that column by exactly $1/\hat{\mu}_{ij}$, the same scalar assigned to it by $(\Lambda_i^R)^{-1/2}$. This holds column by column, so

$$\Lambda_i^{-1/2} R_i = R_i (\Lambda_i^R)^{-1/2}.$$

Post-multiplying by $R_i^\top$ and using the complementary projector identity $R_i R_i^\top = I_{N_i} - P_i P_i^\top$ from Section 6.2:

$$R_i (\Lambda_i^R)^{-1/2} R_i^\top = \Lambda_i^{-1/2} R_i R_i^\top = \Lambda_i^{-1/2} (I_{N_i} - P_i P_i^\top),$$

which is (10). Substituting into the expression above gives the stated result.

Corollary 1 (Consistency with the single-image case) *When $n_i = 1$, Proposition 8 reduces exactly to (7).*

With $n_i = 1$: $P_i = \mathbf{1}_{N_i}/\sqrt{N_i}$, so $P_i P_i^\top = \frac{1}{N_i} \mathbf{1} \mathbf{1}^\top$, and $\Lambda_i^{-1/2} = \hat{\mu}_i^{-1} I_{N_i}$ (constant, one image). The reduced matrix is the scalar $\bar{G}_i = \hat{\mu}_i^2 (1 - \alpha_i N_i)$ (Section 6.5), with trivial eigendecomposition $\bar{V}_i = [1]$, $\bar{\Lambda}_i^{\text{eig}} = \hat{\mu}_i^2 (1 - \alpha_i N_i)$. Substituting:

$$G_i^{-1/2} = \frac{1}{\hat{\mu}_i} \left(I_{N_i} - \frac{1}{N_i} \mathbf{1} \mathbf{1}^\top \right) + \frac{1}{\hat{\mu}_i \sqrt{1 - \alpha_i N_i}} \cdot \frac{1}{N_i} \mathbf{1} \mathbf{1}^\top = \frac{1}{\hat{\mu}_i} I_{N_i} + \frac{1}{\hat{\mu}_i N_i} \left(\frac{1}{\sqrt{1 - \alpha_i N_i}} - 1 \right) \mathbf{1} \mathbf{1}^\top,$$

which is exactly (7).

6.8 The CR2 Adjustment Matrix A_i (General Case)

Applying $A_i = D_i G_i^{-1/2} D_i$ to Proposition 8 requires passing D_i through P_i . Write $\bar{D}_i = \text{diag}(\hat{\mu}_{i1}^{1/2}, \dots, \hat{\mu}_{i,n_i}^{1/2}) \in R^{n_i \times n_i}$ for the image-level analogue of D_i ; since D_i is constant on each image block, the same argument used for $\Lambda_i^{-1/2}$ in the proof of Proposition 8 gives

$$D_i P_i = P_i \bar{D}_i, \quad P_i^\top D_i = \bar{D}_i P_i^\top.$$

First term. Using $\Lambda_i^{-1/2} = D_i^{-2}$ (since $\Lambda_i = \text{diag}(\boldsymbol{\mu}_i^2) = V_i^2 = D_i^4$):

$$D_i \Lambda_i^{-1/2} D_i = D_i D_i^{-2} D_i = I_{N_i}.$$

For the remaining piece, push D_i through P_i on each side, using $\Lambda_i^{-1/2} P_i = P_i \bar{\Lambda}_i^{-1/2}$ (same argument again, with $\bar{\Lambda}_i^{-1/2} = \text{diag}(\hat{\mu}_{i1}^{-1}, \dots, \hat{\mu}_{i,n_i}^{-1})$) and the identity above:

$$D_i \Lambda_i^{-1/2} P_i P_i^\top D_i = (D_i P_i) \bar{\Lambda}_i^{-1/2} (P_i^\top D_i) = P_i (\bar{D}_i \bar{\Lambda}_i^{-1/2} \bar{D}_i) P_i^\top = P_i P_i^\top,$$

since $\bar{D}_i \bar{\Lambda}_i^{-1/2} \bar{D}_i = I_{n_i}$ by the same diagonal cancellation as above, applied at the image level. Hence

$$D_i \Lambda_i^{-1/2} (I_{N_i} - P_i P_i^\top) D_i = I_{N_i} - P_i P_i^\top.$$

Second term. Pushing D_i through P_i on each side of the second term of Proposition 8:

$$D_i \left[P_i \bar{V}_i (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \bar{V}_i^\top P_i^\top \right] D_i = P_i \left(\bar{D}_i \bar{V}_i (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \bar{V}_i^\top \bar{D}_i \right) P_i^\top.$$

Unlike the first term, this does not simplify further: $\bar{D}_i$ and $\bar{V}_i, \bar{\Lambda}_i^{\text{eig}}$ arise from different sources (the fitted means directly, versus the reduced eigendecomposition of Section 6.6) and do not cancel.

Assembling A_i . Adding both terms:

$$A_i = I_{N_i} + P_i \left(\bar{D}_i \bar{V}_i (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \bar{V}_i^\top \bar{D}_i - I_{n_i} \right) P_i^\top, \quad (11)$$

where $\bar{D}_i = \text{diag}(\hat{\mu}_{i1}^{1/2}, \dots, \hat{\mu}_{i,n_i}^{1/2})$, and $\bar{V}_i, \bar{\Lambda}_i^{\text{eig}}$ are the eigenvectors and eigenvalues of the reduced matrix $\bar{G}_i$ (Section 6.5), obtained via `dpr1eig` (Section 6.6). The adjustment matrix is thus the identity plus a correction of rank at most n_i , confined entirely to the image-indicator subspace $\text{range}(P_i)$; the correction itself is characterised by the $n_i \times n_i$ quantities $\bar{D}_i$, $\bar{V}_i$, and $\bar{\Lambda}_i^{\text{eig}}$, and no $N_i \times N_i$ matrix is formed or eigen-decomposed at any point.

Corollary 2 When $n_i = 1$, (11) reduces exactly to (8).

With $n_i = 1$: $\bar{D}_i = \hat{\mu}_i^{1/2}$, $\bar{V}_i = [1]$, $\bar{\Lambda}_i^{\text{eig}} = \hat{\mu}_i^2(1 - \alpha_i N_i)$ (Section 6.5), so the correction term is the scalar

$$\hat{\mu}_i^{1/2} (\hat{\mu}_i^2 (1 - \alpha_i N_i))^{-1/2} \hat{\mu}_i^{1/2} - 1 = \frac{1}{\sqrt{1 - \alpha_i N_i}} - 1,$$

and $P_i(\cdot)P_i^\top = (\cdot) \cdot \frac{1}{N_i} \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top$, recovering (8).

6.9 Complexity

Equation (11) makes the cost of applying A_i to a vector immediate. Write

$$\bar{A}_i := \bar{D}_i \bar{V}_i \left(\bar{\Lambda}_i^{\text{eig}} \right)^{-1/2} \bar{V}_i^\top \bar{D}_i - I_{n_i} \in R^{n_i \times n_i};$$

then for $\hat{r}_i \in R^{N_i}$,

$$A_i \hat{r}_i = \hat{r}_i + P_i \bar{A}_i (P_i^\top \hat{r}_i).$$

Computing $P_i^\top \hat{r}_i$ — a weighted sum of $\hat{r}_i$ within each image — costs $O(N_i)$, since each of the N_i cells contributes to exactly one entry of the result. Forming $\bar{A}_i$ requires the reduced eigenpairs $\bar{V}_i, \bar{\Lambda}_i^{\text{eig}}$, obtained via `dpr1eig` in $O(n_i^2)$ (Section 6.6), together with $O(n_i^2)$ for the surrounding matrix products with $\bar{D}_i$; the subsequent matrix-vector product $\bar{A}_i (P_i^\top \hat{r}_i)$ costs a further $O(n_i^2)$. Broadcasting the result back to all N_i cells via $P_i(\cdot)$ and adding to $\hat{r}_i$ costs $O(N_i)$. Summing,

$$A_i \hat{r}_i \text{ costs } O(N_i + n_i^2).$$

This compares to $O(N_i^3)$ for forming G_i explicitly as an $N_i \times N_i$ matrix and computing its eigendecomposition via a generic dense symmetric eigensolver — the approach taken by the current `clubSandwich` R package implementation of the BRL adjustment (Pustejovsky and Tipton, 2018) — the standard cost of which also governs reconstructing $G_i^{-1/2}$, and hence A_i , as a dense matrix. The gain is substantial whenever $n_i \ll N_i$ — the typical regime for `spicyGLM`, where a patient’s images each contain many reference cells. It narrows as n_i approaches N_i (many images with few cells each), a regime not typical of `spicyGLM`’s design.

When $n_i = 1$, this reduces to direct evaluation of $A_i \hat{r}_i$ via the closed form (8), recovering Algorithm 1 of Section 5.

6.10 Computational Algorithm

Algorithm 3 Closed-Form CR2 Adjustment (Multiple Images Per Patient)

Require: Fitted `spicyGLM`, fitted means $\{\hat{\mu}_{ij}\}$, cells per image $\{n_{ij}\}$, group membership $\{g(i)\}$, residual vectors $\{\hat{r}_i\}$

Ensure: $A_i \hat{r}_i$ for each patient i , via $A_i = I_{N_i} + P_i \bar{A}_i P_i^\top$ (eq. (11))

- 1: $S_g \leftarrow \sum_{i: g(i)=g} T_i$, $T_i = \sum_j n_{ij} \hat{\mu}_{ij}$, for $g \in \{1, 2\}$ $\triangleright O(N)$
 - 2: **for** each patient $i = 1, \dots, m$ **do**
 - 3: $\bar{\Lambda}_i \leftarrow \text{diag}(\hat{\mu}_{i1}^2, \dots, \hat{\mu}_{i, n_i}^2)$; $(\bar{\mathbf{f}}_i)_j \leftarrow \sqrt{n_{ij}} \hat{\mu}_{ij}^{3/2}$ $\triangleright$ Proposition 6, $O(n_i)$
 - 4: $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i} \leftarrow$ Algorithm 2 applied to $\bar{\Lambda}_i, \bar{\mathbf{f}}_i, S_{g(i)}$ $\triangleright O(n_i^2)$
 - 5: $\bar{D}_i \leftarrow \text{diag}(\hat{\mu}_{i1}^{1/2}, \dots, \hat{\mu}_{i, n_i}^{1/2})$
 - 6: $\bar{A}_i \leftarrow \bar{D}_i \bar{V}_i (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \bar{V}_i^\top \bar{D}_i - I_{n_i}$ $\triangleright O(n_i^2)$
 - 7: $d \leftarrow P_i^\top \hat{r}_i$ $\triangleright$ image-wise weighted sums, $O(N_i)$
 - 8: $A_i \hat{r}_i \leftarrow \hat{r}_i + P_i (\bar{A}_i d)$ $\triangleright O(N_i + n_i^2)$
 - 9: **end for**
-

When $n_i = 1$ for all patients, Algorithm 3 reduces to Algorithm 1.

6.11 The `dpr1eig` Algorithm, Adapted to Our Notation

For completeness and self-containedness, this section reproduces the `dpr1eig` algorithm of Jakovčević Stor et al. (2015, Algorithm 1), translated into the notation of this thesis. It computes a single eigenpair $(\bar{\lambda}_k, \bar{v}_k)$ of the reduced matrix $\bar{G}_i$ (after the sign flip of Section 6.6, so here $D = -\bar{\Lambda}_i$, $z = \bar{\mathbf{f}}_i$, $\rho = 1/S_{g(i)} > 0$, with poles $d_1 > \dots > d_{n_i}$ the sorted entries of $-\bar{\Lambda}_i$ and ζ_j the correspondingly reordered entries of $\bar{\mathbf{f}}_i$); the sign is undone and the result relabelled λ_k, v_k upon return, as in Algorithm 2.

Algorithm 4: `dpr1eig`(D, z, ρ, k) — k -th eigenpair of an ordered irreducible DPR1 matrix $A = D + \rho z z^\top$, $\rho > 0$, adapted from Jakovčević Stor et al. (2015, Algorithm 1)

Require: Sorted poles $d_1 > \dots > d_n$, vector $z = (\zeta_1, \dots, \zeta_n)$, scalar $\rho > 0$, index k

Ensure: Eigenpair (λ, v)

```

1: if  $k = 1$  then
2:    $\sigma \leftarrow d_1$  ▷ exterior eigenvalue
3: else
4:    $\bar{D} \leftarrow D - d_k$ ;    $\tau \leftarrow \bar{D}_{k-1}/2$ 
5:    $F \leftarrow 1 + \rho \sum_j \zeta_j^2 / (\bar{D}_j - \tau)$ 
6:   if  $F > 0$  then
7:      $\sigma \leftarrow d_k$ 
8:   else
9:      $\sigma \leftarrow d_{k-1}$ 
10:  end if
11: end if
12: Form the shifted matrix  $A_i = A - \sigma I$ , split into blocks  $D_1$  (poles above  $\sigma$ ),  $D_2$  (poles below  $\sigma$ ),  $z_1, z_2$  (corresponding entries of  $z$ ), and  $\zeta_i$  (entry at  $\sigma$ )
13: Compute  $w_1 = -D_1^{-1} z_1 / \zeta_i$ ,  $w_2 = -D_2^{-1} z_2 / \zeta_i$ , and the pivot entry  $b$  (eq. 7–8 of the cited paper; see Remark 5 below)
14: Compute  $K_b, K_z, \kappa_\nu$  (Remark 5); if  $\kappa_\nu \gg n$ , recompute  $b$  in extended precision
15: if  $\sigma = d_{k-1}$  then
16:   Compute the leftmost eigenvalue  $\nu$  of  $A_i^{-1}$  (via the secular equation on the arrowhead structure, eq. 9)
17: else
18:   Compute the rightmost eigenvalue  $\nu$  of  $A_i^{-1}$ 
19: end if
20:  $\mu \leftarrow 1/\nu$ ; compute  $v$  via eq. 10 of the cited paper;  $\lambda \leftarrow \mu + \sigma$ 
21: Compute  $K_\nu$  (Remark 5); if  $K_\nu \gg 1$ , apply remedy R1 (shift to the other neighbouring pole) or, if unavailable, R2 (shift to an arbitrary nearby  $\sigma$  strictly between  $\lambda$  and its neighbouring poles), as described in the cited paper
22: if  $|\lambda| \ll \min\{|\lambda - d_k|, |\lambda - d_{k-1}|\}$  then
23:   Recompute  $\lambda$  from  $A^{-1}$  directly (eq. 13) to avoid cancellation
24: end if
25: return  $(\lambda, v)$ 

```

This is invoked once per $k = 1, \dots, n_i$ in Algorithm 2 to obtain the full eigendecomposition of $\bar{G}_i$.

Remark 5 (Precision diagnostics) *The pivot entry b is computed as*

$$b = \frac{1}{\zeta_i^2} \left(\frac{1}{\rho} + z_1^\top D_1^{-1} z_1 + z_2^\top D_2^{-1} z_2 \right).$$

Its reliability is governed by

$$K_b = \frac{|1 + \rho z_1^\top D_1^{-1} z_1 - \rho z_2^\top D_2^{-1} z_2|}{|1 + \rho z_1^\top D_1^{-1} z_1 + \rho z_2^\top D_2^{-1} z_2|}, \quad K_z = \frac{1}{|\zeta_i|} \sum_{j \neq i} |\zeta_j|,$$

$$\kappa_\nu \leq \min \{ (n+4)\sqrt{n} K_b, 3\sqrt{n} + (n+4)(1+2K_z) \} \quad (\text{eq. 11}),$$

and the choice of shift pole is validated by

$$K_\nu = \frac{\|A_i^{-1}\|_2}{|\nu|} \quad (\text{eq. 12}).$$

See Jakovčević Stor et al. (2015, Sections 2.1–2.3) for the derivation and interpretation of these quantities.

6.12 Solving the Reduced Problem via Direct Eigendecomposition

An alternative to `dpr1eig` (Section 6.6) is to solve the reduced eigenproblem $\bar{G}_i \bar{v} = \bar{\lambda} \bar{v}$ directly, by forming $\bar{G}_i$ explicitly as a dense $n_i \times n_i$ matrix via Proposition 6 and passing it to a general-purpose symmetric eigensolver. We refer to this as the *direct route*, in contrast to `dpr1eig`'s *shift-and-invert route*.

Algorithm 5 Reduced eigendecomposition of $\bar{G}_i$ via direct eigendecomposition

Require: Fitted means $\{\hat{\mu}_{ij}\}_{j=1}^{n_i}$, number of cells per image $\{n_{ij}\}_{j=1}^{n_i}$, group total $S_{g(i)}$

Ensure: Eigenpairs $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i}$ of $\bar{G}_i$

- 1: $\bar{\Lambda}_i \leftarrow \text{diag}(\hat{\mu}_{i1}^2, \dots, \hat{\mu}_{i,n_i}^2); \quad (\bar{\mathbf{f}}_i)_j \leftarrow \sqrt{n_{ij}} \hat{\mu}_{ij}^{3/2}$ ▷ Proposition 6, $O(n_i)$
 - 2: $\bar{G}_i \leftarrow \bar{\Lambda}_i - \frac{1}{S_{g(i)}} \bar{\mathbf{f}}_i \bar{\mathbf{f}}_i^\top$ ▷ form explicitly, $O(n_i^2)$
 - 3: $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i} \leftarrow \text{eigen}(\bar{G}_i)$ ▷ generic symmetric eigensolver, $O(n_i^3)$
 - 4: **return** $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i}$
-

Unlike Algorithm 2, this route requires none of the machinery specific to the DPR1 structure. Three simplifications follow directly:

Remark 6 (No sign flip required) *`dpr1eig` requires $\rho > 0$ (Section 6.6), forcing the decomposition of $-\bar{G}_i$ rather than $\bar{G}_i$ directly, with eigenvalues negated on return. A generic symmetric eigensolver imposes no such constraint: $\bar{G}_i$ is passed and returned as-is, eliminating this bookkeeping step entirely.*

Remark 7 (Tied poles require no special handling) *`dpr1eig` requires distinct diagonal entries of D (irreducibility; Section 6.6), and exact ties among $\hat{\mu}_{ij}^2$ must be detected*

and handled via the Givens-rotation-based reduction of Remark 4 before the shift-and-invert machinery can be applied. A generic symmetric eigensolver imposes no such restriction: repeated eigenvalues are a routine, numerically stable case for standard dense eigensolvers (e.g. the LAPACK routine `dsyevr`, called internally by R’s `eigen()`), and Algorithm 5 requires no tie-detection step of any kind.

Corollary 3 (The $n_i = 1$ case is trivial, requiring no guard) *When $n_i = 1$, $\bar{G}_i$ is a 1×1 matrix and Algorithm 5 returns $\bar{\lambda}_1 = \bar{G}_i$, $\bar{v}_1 = 1$ directly, with no special-casing required. This contrasts with `dpr1eig` (Algorithm 4), whose secular-equation machinery is undefined at $n = 1$ and requires an explicit boundary case returning $\lambda = d_1 + \rho\zeta_1^2$, $v = 1$ directly (consistent with Corollary 1 by the same reasoning as Section 5.6).*

Algorithm 5 substitutes directly for Algorithm 2 within Algorithm 3 (Section 6.10): the eigenpairs $(\bar{\lambda}_k, \bar{v}_k)_{k=1}^{n_i}$ it produces feed into the assembly of $\bar{A}_i$ (eq. (11)) identically, regardless of which route produced them.

Complexity. Algorithm 5 costs $O(n_i^3)$, versus `dpr1eig`’s $O(n_i^2)$ (Section 6.6) — a worse asymptotic exponent. This does not alter the overall complexity claim of Section 6.9: since $n_i \ll N_i$ in the spicyGLM regime, $O(N_i + n_i^3)$ remains negligible relative to `clubSandwich`’s $O(N_i^3)$ dense approach whenever $n_i \ll N_i^{2/3}$, which holds throughout the practical range of images per patient. The comparison between Algorithms 2 and 5 is therefore an empirical, not asymptotic, question, addressed in Section 8.

7 Implementation Plan (Overview)

Ideally, a standalone R/C implementation is planned, validated against but not wrapped around `clubSandwich`, avoiding the internal `v_scale/w_scale` reconciliation that a wrapper would require.

Both routes to $\bar{G}_i$ ’s eigendecomposition will be implemented and empirically compared: (a) generic `eigen()`, and (b) `dpr1eig` (Section 6.6). Since $\bar{G}_i$ is only $n_i \times n_i$ with $n_i \ll N_i$, it is not obvious a priori which is faster in practice — `eigen()`’s $O(n_i^3)$ cost may be negligible at this scale despite its worse asymptotic exponent relative to `dpr1eig`’s $O(n_i^2)$, so the two will be timed head-to-head across a range of n_i rather than one being assumed superior. Correctness will be established and validated against `clubSandwich` as the accuracy benchmark, before benchmarking both eigendecomposition routes and the overall asymptotic speedup of Section 6.9.

8 Empirical Comparison of Eigendecomposition Routes

Section 6.12 noted that the worse asymptotic exponent of the direct route (Algorithm 5, $O(n_i^3)$) relative to `dpr1eig` (Algorithm 2, $O(n_i^2)$) need not translate into worse practical performance, since $\bar{G}_i$ is small ($n_i \ll N_i$) throughout the spicyGLM regime. We resolve

this empirically by timing both routes head-to-head on well-conditioned synthetic DPR1 problems across a range of n_i .

8.1 Method

For each $n_i \in \{2, 5, 10, 20, 50, 100, 200\}$, a synthetic DPR1 problem $(\bar{\Lambda}_i, \bar{\mathbf{f}}_i, \rho)$ was generated with distinct, well-separated poles (drawn uniformly on $[1, 10]$) and positive weights (drawn uniformly on $[0.5, 2]$), $\rho = 1$. Each route was timed over 30 repetitions on the identical generated problem via median wall-clock time.

8.2 Results

n_i	<code>eigen()</code> (μs)	<code>dpr1eig</code> (μs)	ratio
2	21.1	447.2	$21\times$
5	27.5	1017.6	$37\times$
10	40.6	1929.9	$48\times$
20	78.3	4413.2	$56\times$
50	497.0	12451.3	$25\times$
100	2478.9	26228.7	$11\times$
200	15328.0	88312.6	$6\times$

Table 3: Median wall-clock time (microseconds, $n = 30$) for the direct route (`eigen()`) versus `dpr1eig`, on well-conditioned synthetic DPR1 problems.

The direct route is faster at every tested n_i , by a factor of 6–56 $\times$, with the largest margins at small n_i and a narrowing (though still substantial) gap as n_i grows. This is consistent with Jakovčević Stor et al. (2015, Table 2)’s own reported timings, in which their optimized Julia implementation of `dpr1eig` was itself 8–17 $\times$ slower than the compiled LAPACK routine `dlaed9` at $n = 2002$: the overhead is attributable to R’s interpreted-language function-call and iteration cost in the shift-and-invert route (bracket search, bisection, per-eigenpair looping), which compiled dense eigensolvers do not incur, and which dominates for problems of the size $\bar{G}_i$ takes in the spicyGLM setting (n_i typically in the single digits, rarely exceeding ~ 20).

9 Degrees of Freedom

This section derives, from first principles, a closed-form Satterthwaite degrees of freedom for the test statistic for spicyGLM. The general strategy — writing the variance estimator as a quadratic form in the true residuals and matching moments via the eigenvalues of an associated Gram matrix — is due to Bell and McCaffrey (2002) and extended to weighted and fixed-effects models by Pustejovsky and Tipton (2018). The specific loading vector for the multi-image Poisson GLM setting, and the closed-form entries of the resulting Gram matrix, are derived below.

9.1 Setup

Let $\varepsilon_i := Y_i - \mu_i(\beta) \in R^{N_i}$ denote patient i 's residual at the true parameter β , so that $\text{Cov}(\varepsilon_i) = V_i = \text{diag}(\hat{\mu}_i)$ and $\varepsilon_i \perp \varepsilon_k$ for $i \neq k$. Let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)$ be the full stacked vector, with block-diagonal covariance $\Sigma = \text{blockdiag}(V_1, \dots, V_m)$. Recall $w := \mathcal{B}L^\top = (-1/S_1, 1/S_2)^\top$, $\tilde{\tau}_j := \mathbf{1}_{N_j}^\top A_j \hat{\mu}_j$, and

$$e_j(\beta') := w_{g(j)} \cdot \mathbf{1}_{N_j}^\top A_j (Y_j - \mu_j(\beta')), \quad \hat{v} := \sum_{j=1}^m e_j(\hat{\beta})^2.$$

9.2 Sensitivity of the Estimating Equation

Lemma 3

$$\frac{\partial \hat{\beta}}{\partial Y_k} = \mathcal{B}X_k^\top, \quad \hat{\beta} - \beta \approx \mathcal{B} \sum_{l=1}^m X_l^\top \varepsilon_l.$$

$\hat{\beta}$ solves $S(\beta) := \sum_{l=1}^m X_l^\top (Y_l - \mu_l(\beta)) = 0$ for every realization of Y . Differentiating this identity with respect to Y_k via the chain rule,

$$0 = \left. \frac{\partial S}{\partial Y_k} \right|_{\beta \text{ fixed}} + \frac{\partial S}{\partial \beta} \cdot \frac{\partial \hat{\beta}}{\partial Y_k} = X_k^\top - \mathcal{I} \cdot \frac{\partial \hat{\beta}}{\partial Y_k}, \quad \mathcal{I} := \sum_{l=1}^m X_l^\top V_l X_l,$$

using $\partial \mu_l / \partial \beta = V_l X_l$ (Section 4). Solving gives $\partial \hat{\beta} / \partial Y_k = \mathcal{I}^{-1} X_k^\top = \mathcal{B}X_k^\top$. Summing each cluster's marginal contribution to a first-order Taylor expansion of $\hat{\beta}$ about β gives the second identity stated above.

9.3 The Loading Vector g_j

The linearization of $e_j(\hat{\beta})$ derived below is a weighted sum of every patient's true residual $\varepsilon_1, \dots, \varepsilon_m$, not just ε_j . Since $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)$ is itself a single stacked vector, this weighted sum can be written compactly as an inner product $g_j^\top \varepsilon$ for some single vector g_j , of the same length as ε , whose l -th block, $[g_j]_l \in R^{N_l}$, records the weight applied to patient l 's residual. We refer to g_j as patient j 's *loading vector*: it summarizes, in one object, how $e_j(\hat{\beta})$ responds to every patient's data.

Lemma 4 *To first order,*

$$e_j(\hat{\beta}) \approx g_j^\top \varepsilon, \quad [g_j]_l = \begin{cases} w_{g(j)} A_j \mathbf{1}_{N_j} - w_{g(j)} \frac{\tilde{\tau}_j}{S_{g(j)}} \mathbf{1}_{N_j}, & l = j, \\ -w_{g(j)} \frac{\tilde{\tau}_j}{S_{g(j)}} \mathbf{1}_{N_l}, & l \neq j, \ g(l) = g(j), \\ 0, & g(l) \neq g(j). \end{cases}$$

Differentiating $e_j(\beta')$ at $\beta' = \beta$, treating A_j as fixed (the same first-order treatment used in the derivation of A_i itself, Section 4):

$$\left. \frac{\partial e_j}{\partial \beta'} \right|_{\beta} = -w_{g(j)} \mathbf{1}_{N_j}^\top A_j V_j X_j = -w_{g(j)} \left(\mathbf{1}_{N_j}^\top A_j \hat{\mu}_j \right) \mathbf{e}_{g(j)}^\top = -w_{g(j)} \tilde{\tau}_j \mathbf{e}_{g(j)}^\top,$$

using $X_j = \mathbf{1}_{N_j} \mathbf{e}_{g(j)}^\top$ and $V_j \mathbf{1}_{N_j} = \hat{\mu}_j$. A first-order Taylor expansion of e_j about β , substituting Lemma 3, gives

$$e_j(\hat{\beta}) \approx \underbrace{w_{g(j)} \mathbf{1}_{N_j}^\top A_j \varepsilon_j}_{\text{direct term}} - \underbrace{w_{g(j)} \tilde{\tau}_j \mathbf{e}_{g(j)}^\top \mathcal{B} \sum_{l=1}^m X_l^\top \varepsilon_l}_{\text{correction term}}.$$

The direct term already has the form of one block of an inner product with ε : it contributes $w_{g(j)} A_j \mathbf{1}_{N_j}$ to $[g_j]_j$ and nothing to any other block. The correction term is a sum over every patient $l = 1, \dots, m$. Expanding the correction term into its own m blocks and identifying $[g_j]_l$ as the coefficient of ε_l in each requires evaluating $\mathbf{e}_{g(j)}^\top \mathcal{B} X_l^\top$ for a general l . Using $\mathbf{e}_{g(j)}^\top \mathcal{B} = S_{g(j)}^{-1} \mathbf{e}_{g(j)}^\top$ (since $\mathcal{B}$ is diagonal) and $\mathbf{e}_{g(j)}^\top X_l^\top = \mathbf{e}_{g(j)}^\top \mathbf{e}_{g(l)} \mathbf{1}_{N_l}^\top = 1[g(j) = g(l)] \mathbf{1}_{N_l}^\top$ (since $\mathbf{e}_{g(j)}$ and $\mathbf{e}_{g(l)}$ are orthogonal standard basis vectors unless $g(j) = g(l)$), the correction term's contribution to block l is

$$-w_{g(j)} \tilde{\tau}_j S_{g(j)}^{-1} 1[g(j) = g(l)] \mathbf{1}_{N_l}.$$

Adding this to the direct term at $l = j$, and taking it alone for every $l \neq j$ with $g(l) = g(j)$ (zero otherwise, since the indicator vanishes), gives the three cases of $[g_j]_l$ stated above.

9.4 Exact Distribution of $\hat{v}$

Let $M := \sum_{l=1}^m N_l$ denote the total number of reference cells across all patients, so that $\varepsilon \in R^M$ and each $g_j \in R^M$ (Lemma 4).

Proposition 9 *Writing $\varepsilon = \Sigma^{1/2} z$ for $z \sim N(0, I_M)$,*

$$\hat{v} \approx \sum_{i=1}^M \lambda_i u_i^2,$$

for u_i i.i.d. $N(0, 1)$, where $\{\lambda_i\}$ are the eigenvalues of the $M \times M$ matrix

$$G^* := \Sigma^{1/2} \left(\sum_{j=1}^m g_j g_j^\top \right) \Sigma^{1/2}.$$

By Lemma 4, $e_j(\hat{\beta}) \approx g_j^\top \varepsilon$, so

$$\hat{v} = \sum_{j=1}^m e_j(\hat{\beta})^2 \approx \sum_{j=1}^m (g_j^\top \varepsilon)^2 = \sum_{j=1}^m \varepsilon^\top g_j g_j^\top \varepsilon = \varepsilon^\top \left(\sum_{j=1}^m g_j g_j^\top \right) \varepsilon,$$

using $(g_j^\top \varepsilon)^2 = (g_j^\top \varepsilon)(\varepsilon^\top g_j) = \varepsilon^\top (g_j g_j^\top) \varepsilon$ for each j . Substituting $\varepsilon = \Sigma^{1/2} z$ and using $(\Sigma^{1/2})^\top = \Sigma^{1/2}$,

$$\hat{v} \approx z^\top \Sigma^{1/2} \left(\sum_j g_j g_j^\top \right) \Sigma^{1/2} z = z^\top G^* z.$$

G^* is symmetric, so it admits an eigendecomposition $G^* = U\Lambda U^\top$ with U orthogonal ($U^\top U = I_M$) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_M)$. Let $y := U^\top z$; since U is orthogonal and $z \sim N(0, I_M)$, $\text{Cov}(y) = U^\top \text{Cov}(z)U = U^\top U = I_M$, so $y = (u_1, \dots, u_M)$ with u_i i.i.d. $N(0, 1)$. Then

$$z^\top G^* z = z^\top U\Lambda U^\top z = (U^\top z)^\top \Lambda (U^\top z) = y^\top \Lambda y = \sum_{i=1}^M \lambda_i u_i^2.$$

Computing the M eigenvalues of G^* directly would require forming and diagonalizing an $M \times M$ matrix, at cost comparable to the dense computation that the entire fast estimator is designed to avoid. The following proposition shows that the same nonzero eigenvalues can instead be obtained from an $m \times m$ matrix, with m the number of patients.

Proposition 10 (Gram-matrix reduction) *Let $B := \Sigma^{1/2}[g_1 \ \dots \ g_m] \in R^{M \times m}$, so that $G^* = BB^\top$. Define $P := B^\top B \in R^{m \times m}$, with entries*

$$P_{jk} = g_j^\top \Sigma g_k.$$

Every nonzero eigenvalue of $G^ = BB^\top$ is an eigenvalue of $P = B^\top B$, with the same multiplicity, and vice versa.*

$P_{jk} = (B^\top B)_{jk} = (\Sigma^{1/2}g_j)^\top (\Sigma^{1/2}g_k) = g_j^\top \Sigma g_k$, confirming the stated entries. Suppose $\lambda \neq 0$ and $G^*v = \lambda v$ for some $v \neq 0$, i.e. $BB^\top v = \lambda v$. Then

$$P(B^\top v) = B^\top B(B^\top v) = B^\top (BB^\top v) = B^\top (\lambda v) = \lambda(B^\top v),$$

so $B^\top v$ is an eigenvector of P with eigenvalue λ , provided $B^\top v \neq 0$. If $B^\top v = 0$, then $G^*v = BB^\top v = B \cdot 0 = 0$, contradicting $\lambda \neq 0$; hence $B^\top v \neq 0$. This shows every nonzero eigenvalue of G^* is an eigenvalue of P ; exchanging the roles of B and $B^\top$ shows the reverse inclusion.

Proposition 11 (Degrees of freedom via traces)

$$\hat{\nu} = \frac{\left(\sum_j P_{jj}\right)^2}{\sum_{j,k} P_{jk}^2}.$$

For $u \sim N(0, 1)$, $E[u^2] = 1$ and $E[u^4] = 3$, so $\text{Var}(u^2) = E[u^4] - (E[u^2])^2 = 2$. By Propositions 9 and 10, the nonzero λ_i are the eigenvalues of P , so, using independence of the u_i ,

$$E[\hat{\nu}] = \sum_i \lambda_i = \text{tr}(P) = \sum_j P_{jj}, \quad \text{Var}(\hat{\nu}) = 2 \sum_i \lambda_i^2 = 2(P^2) = 2 \sum_{j,k} P_{jk}^2,$$

the last equality using $(P^2) = \sum_j (P^2)_{jj} = \sum_j \sum_k P_{jk} P_{kj} = \sum_{j,k} P_{jk}^2$ (symmetry, $P_{kj} = P_{jk}$). Matching these to the mean $c\hat{\nu}$ and variance $2c^2\hat{\nu}$ of a scaled $\chi_{\hat{\nu}}^2$ variable $c\chi_{\hat{\nu}}^2$, and dividing the square of the mean equation by the variance equation to eliminate c ,

$$\hat{\nu} = \frac{(c\hat{\nu})^2}{c^2\hat{\nu}} = \frac{\left(\sum_j P_{jj}\right)^2}{\sum_{j,k} P_{jk}^2}.$$

9.5 Closed Form of P_{jk}

Since Σ is block-diagonal with blocks $V_1, \dots, V_m$,

$$P_{jk} = g_j^\top \Sigma g_k = \sum_{l=1}^m [g_j]_l^\top V_l [g_k]_l,$$

a sum over patients l of the pairing of g_j 's and g_k 's l -th blocks (Lemma 4).

For $g(j) \neq g(k)$, $P_{jk} = 0$. For $g(j) = g(k) = g$ and $j \neq k$,

$$P_{jk} = -\frac{\tilde{\tau}_j \tilde{\tau}_k}{S_g^3} = -h_j h_k, \quad h_j := \frac{\tilde{\tau}_j}{S_{g(j)}^{3/2}}.$$

For $j = k$,

$$P_{jj} = P_j^{\text{diag}} - h_j^2, \quad P_j^{\text{diag}} := w_{g(j)}^2 (A_j \mathbf{1}_{N_j})^\top V_j (A_j \mathbf{1}_{N_j}).$$

Different groups. By Lemma 4, $[g_j]_l = 0$ unless $g(l) = g(j)$, and $[g_k]_l = 0$ unless $g(l) = g(k)$. If $g(j) \neq g(k)$, no l satisfies both, so every term in the sum vanishes and $P_{jk} = 0$.

Same group, $j \neq k$. Write $g := g(j) = g(k)$, $w := w_g$. Only patients $l \in g$ contribute. Split the sum at $l = j$, $l = k$, and $l \in g \setminus \{j, k\}$.

At $l = j$: $[g_j]_j = w A_j \mathbf{1}_{N_j} - w \frac{\tilde{\tau}_j}{S_g} \mathbf{1}_{N_j}$ and $[g_k]_j = -w \frac{\tilde{\tau}_k}{S_g} \mathbf{1}_{N_j}$ (patient j is not patient k 's own block). Using $(A_j \mathbf{1}_{N_j})^\top V_j \mathbf{1}_{N_j} = \mathbf{1}_{N_j}^\top V_j A_j \mathbf{1}_{N_j} = \tilde{\tau}_j$ (as $A_j^\top = A_j$, Proposition 1) and $\mathbf{1}_{N_j}^\top V_j \mathbf{1}_{N_j} = T_j$:

$$[g_j]_j^\top V_j [g_k]_j = -w^2 \frac{\tilde{\tau}_k}{S_g} \tilde{\tau}_j + w^2 \frac{\tilde{\tau}_j \tilde{\tau}_k}{S_g^2} T_j = w^2 \tilde{\tau}_j \tilde{\tau}_k \left(-\frac{1}{S_g} + \frac{T_j}{S_g^2} \right).$$

At $l = k$: by the identical computation with j, k exchanged,

$$[g_j]_k^\top V_k [g_k]_k = w^2 \tilde{\tau}_j \tilde{\tau}_k \left(-\frac{1}{S_g} + \frac{T_k}{S_g^2} \right).$$

At each $l \in g \setminus \{j, k\}$: $[g_j]_l = -w \frac{\tilde{\tau}_j}{S_g} \mathbf{1}_{N_l}$, $[g_k]_l = -w \frac{\tilde{\tau}_k}{S_g} \mathbf{1}_{N_l}$, so

$$[g_j]_l^\top V_l [g_k]_l = w^2 \frac{\tilde{\tau}_j \tilde{\tau}_k}{S_g^2} T_l.$$

Summing over $l \in g \setminus \{j, k\}$ and using $\sum_{l \in g} T_l = S_g$:

$$\sum_{l \in g \setminus \{j, k\}} [g_j]_l^\top V_l [g_k]_l = w^2 \frac{\tilde{\tau}_j \tilde{\tau}_k}{S_g^2} (S_g - T_j - T_k).$$

Adding all three pieces:

$$P_{jk} = w^2 \tilde{\tau}_j \tilde{\tau}_k \left[-\frac{2}{S_g} + \frac{T_j + T_k}{S_g^2} + \frac{S_g - T_j - T_k}{S_g^2} \right] = w^2 \tilde{\tau}_j \tilde{\tau}_k \left(-\frac{1}{S_g} \right),$$

since the T_j, T_k terms cancel exactly. Using $w^2 = w_{g(j)}w_{g(k)} = 1/S_g^2$:

$$P_{jk} = -\frac{\tilde{\tau}_j \tilde{\tau}_k}{S_g^3} = -h_j h_k.$$

$j = k$. Split $P_{jj} = \sum_l [g_j]_l^\top V_l [g_j]_l$ at $l = j$ and $l \in g \setminus \{j\}$. At $l = j$:

$$[g_j]_j^\top V_j [g_j]_j = w^2 (A_j \mathbf{1}_{N_j})^\top V_j (A_j \mathbf{1}_{N_j}) - 2w^2 \frac{\tilde{\tau}_j}{S_g} (A_j \mathbf{1}_{N_j})^\top V_j \mathbf{1}_{N_j} + w^2 \frac{\tilde{\tau}_j^2}{S_g^2} \mathbf{1}_{N_j}^\top V_j \mathbf{1}_{N_j},$$

which, using the identities above, equals $P_j^{\text{diag}} - 2w^2 \tilde{\tau}_j^2 / S_g + w^2 \tilde{\tau}_j^2 T_j / S_g^2$. At each $l \neq j$ in g : $[g_j]_l^\top V_l [g_j]_l = w^2 (\tilde{\tau}_j^2 / S_g^2) T_l$, summing to $w^2 (\tilde{\tau}_j^2 / S_g^2) (S_g - T_j)$. Adding both pieces:

$$P_{jj} = P_j^{\text{diag}} + w^2 \tilde{\tau}_j^2 \left[-\frac{2}{S_g} + \frac{T_j}{S_g^2} + \frac{S_g - T_j}{S_g^2} \right] = P_j^{\text{diag}} - w^2 \frac{\tilde{\tau}_j^2}{S_g},$$

the T_j terms again cancelling exactly. Using $w^2 = w_{g(j)}^2 = 1/S_g^2$:

$$P_{jj} = P_j^{\text{diag}} - \frac{\tilde{\tau}_j^2}{S_g^3} = P_j^{\text{diag}} - h_j^2.$$

Corollary 4 *Substituting Theorem 9.5 into Proposition 11:*

$$\hat{\nu} = \frac{\left(\sum_{j=1}^m (P_j^{\text{diag}} - h_j^2) \right)^2}{\sum_{j,k: g(j)=g(k)} (h_j h_k)^2}.$$

The sum in the denominator is restricted to same-group pairs (j, k) , including $j = k$, since $P_{jk} = 0$ for $g(j) \neq g(k)$ (Theorem 9.5), whereas $h_j h_k$ itself need not vanish across groups.

This is exactly the degrees-of-freedom formula used in the implementation (Algorithm 6 below): P_j^{diag} and $\tilde{\tau}_j$ each require one call to Algorithm 3's fast-trick evaluation of $A_j \hat{r}_i$, applied to $\mathbf{1}_{N_j}$ and $\hat{\mu}_j$ respectively in place of $\hat{r}_j$, at the same $O(N_j + n_j^2)$ cost as the variance estimator itself. No dense $N_j \times N_j$ matrix, and no $m \times m$ eigendecomposition, is ever formed.

9.6 Computational Algorithm

Algorithm 6 Satterthwaite Degrees of Freedom

Require: $S_1, S_2, \{g(j)\}, \{\hat{\mu}_j\}$

Ensure: $\hat{\nu}$

- 1: **for** each patient $j = 1, \dots, m$ **do**
 - 2: $P_j^{\text{diag}} \leftarrow w_{g(j)}^2 (A_j \mathbf{1}_{N_j})^\top V_j (A_j \mathbf{1}_{N_j})$ $\triangleright$ via Algorithm 3 applied to $\mathbf{1}_{N_j}$
 - 3: $\tilde{\tau}_j \leftarrow \mathbf{1}_{N_j}^\top A_j \hat{\mu}_j$ $\triangleright$ via Algorithm 3 applied to $\hat{\mu}_j$
 - 4: $h_j \leftarrow \tilde{\tau}_j / S_{g(j)}^{3/2}$
 - 5: **end for**
 - 6: $\hat{\nu} \leftarrow \left(\sum_j (P_j^{\text{diag}} - h_j^2) \right)^2 / \sum_{j,k} (h_j h_k)^2$
 - 7: **return** $\hat{\nu}$
-

10 Firth Correction

10.1 Motivation: Boundary Failure of the MLE

For spicyGLM’s no-intercept, one-parameter-per-group design, the ordinary Poisson MLE is closed-form. Writing

$$Y_g := \sum_{i:g(i)=g} \sum_{j=1}^{n_i} \sum_{k=1}^{n_{ij}} Y_{ijk}, \quad D_g := \sum_{i:g(i)=g} \sum_{j=1}^{n_i} \sum_{k=1}^{n_{ij}} \widehat{\text{den}}_{ij}, \quad S_g(\beta) := \sum_{i:g(i)=g} \sum_{j=1}^{n_i} \sum_{k=1}^{n_{ij}} \mu_{ijk}(\beta) = e^{\beta_g} D_g,$$

the score equation $U_g(\beta) = Y_g - S_g(\beta) = 0$ solves to

$$\hat{\beta}_g^{\text{MLE}} = \log\left(\frac{Y_g}{D_g}\right).$$

Remark 8 (Consistency with earlier notation) $S_g(\beta)$ here is the same quantity as S_g in Table 2 and Section 6.3, written as a function of β rather than evaluated at a fitted $\hat{\mu}$.

This is undefined whenever $Y_g = 0$ — a structural zero, common when a target cell type is entirely absent from a group’s REF-cell neighbourhoods: $\hat{\beta}_g^{\text{MLE}} \rightarrow -\infty$ as $Y_g \rightarrow 0^+$, and no finite MLE exists at $Y_g = 0$ exactly.

10.2 Firth’s Penalized Likelihood: General Derivation

Firth’s (Firth, 1993) bias-reduction device replaces the ordinary log-likelihood $\ell(\beta)$ with the Jeffreys-prior-penalized version

$$\ell^*(\beta) = \ell(\beta) + \frac{1}{2} \log \det \mathcal{I}(\beta),$$

giving the modified score

$$U^*(\beta) = U(\beta) + \frac{1}{2} \frac{\partial}{\partial \beta} \log \det \mathcal{I}(\beta). \quad (12)$$

The penalty gradient is derived from first principles below.

Lemma 5 (Derivative of a log-determinant) *For an invertible, differentiably-parametrised matrix $M(\theta)$,*

$$\frac{\partial}{\partial \theta} \log \det M(\theta) = \text{tr} \left(M(\theta)^{-1} \frac{\partial M(\theta)}{\partial \theta} \right).$$

For a small perturbation ϵdM , $\det(M + \epsilon dM) = \det(M) \det(I + \epsilon M^{-1} dM)$. Since $\det(I + \epsilon A) = 1 + \epsilon \text{tr}(A) + O(\epsilon^2)$ for any matrix A ,

$$\log \det(M + \epsilon dM) - \log \det M = \epsilon \text{tr}(M^{-1} dM) + O(\epsilon^2),$$

and dividing by ϵ and taking $\epsilon \rightarrow 0$ gives the stated derivative.

Recall $\mathcal{I}(\beta) = \sum_{ijk} \mu_{ijk}(\beta) x_{ijk} x_{ijk}^\top$ (Section 3.1), where $\mu_{ijk}(\beta) = V(\mu_{ijk}(\beta))$ for canonical-link exponential families (V the variance function). Applying Lemma 5 with $M(\beta) = \mathcal{I}(\beta)$:

$$\frac{\partial}{\partial \beta_g} \log \det \mathcal{I}(\beta) = \text{tr} \left(\mathcal{I}(\beta)^{-1} \frac{\partial \mathcal{I}(\beta)}{\partial \beta_g} \right). \quad (13)$$

Differentiating $\mathcal{I}(\beta)$ term by term, using the chain rule $\frac{\partial V(\mu_{ijk})}{\partial \beta_g} = V'(\mu_{ijk}) \frac{\partial \mu_{ijk}}{\partial \beta_g}$ and the canonical-log-link identity $\frac{\partial \mu_{ijk}}{\partial \beta_g} = \mu_{ijk} (x_{ijk})_g$ (Section 5.2):

$$\frac{\partial \mathcal{I}(\beta)}{\partial \beta_g} = \sum_{ijk} V'(\mu_{ijk}) \mu_{ijk} (x_{ijk})_g x_{ijk} x_{ijk}^\top.$$

Substituting into (13) and using linearity of the trace together with the scalar identity $\text{tr}(\mathcal{I}(\beta)^{-1} x_{ijk} x_{ijk}^\top) = x_{ijk}^\top \mathcal{I}(\beta)^{-1} x_{ijk}$ (cyclic property of trace, applied to a rank-one matrix):

$$\frac{\partial}{\partial \beta_g} \log \det \mathcal{I}(\beta) = \sum_{ijk} V'(\mu_{ijk}) \underbrace{\mu_{ijk} x_{ijk}^\top \mathcal{I}(\beta)^{-1} x_{ijk}}_{=: h_{ijk}(\beta)} (x_{ijk})_g.$$

Definition 2 (Firth leverage)

$$h_{ijk}(\beta) := \mu_{ijk}(\beta) x_{ijk}^\top \mathcal{I}(\beta)^{-1} x_{ijk}.$$

Hence, in full generality for a canonical-link GLM,

$$\frac{\partial}{\partial \beta_g} \log \det \mathcal{I}(\beta) = \sum_{ijk} V'(\mu_{ijk}) h_{ijk}(\beta) (x_{ijk})_g. \quad (14)$$

Poisson specialization. $V(\mu) = \mu \implies V'(\mu) \equiv 1$, so (14) collapses to

$$\frac{\partial}{\partial \beta_g} \log \det \mathcal{I}(\beta) = \sum_{ijk} h_{ijk}(\beta) (x_{ijk})_g.$$

Substituting into (12) with $U_g(\beta) = \sum_{ijk} (x_{ijk})_g (Y_{ijk} - \mu_{ijk})$:

$$U_g^*(\beta) = \sum_{ijk} (x_{ijk})_g \left(Y_{ijk} - \mu_{ijk}(\beta) + \frac{1}{2} h_{ijk}(\beta) \right). \quad (15)$$

Equation (15) says Firth-corrected Poisson regression is ordinary Poisson estimation run on pseudo-responses $Y_{ijk}^* = Y_{ijk} + h_{ijk}(\beta)/2$: Poisson's linear variance function ($V' \equiv 1$) is what makes the correction a flat additive shift, independent of μ_{ijk} .

10.3 Closed-Form Firth Estimate for spicyGLM

For spicyGLM's design, $x_{ijk} = \mathbf{e}_{g(i)}$ and $\mathcal{I}(\beta) = \text{diag}(S_1(\beta), S_2(\beta))$ (Section 5.2), so Definition 2 specializes to

$$h_{ijk}(\beta) = \frac{\mu_{ijk}(\beta)}{S_{g(i)}(\beta)}.$$

Proposition 12 (Group leverage sums to one, at every β) *For every group $g \in \{1, 2\}$ and every $\beta \in R^2$,*

$$\sum_{i:g(i)=g} \sum_{j=1}^{n_i} \sum_{k=1}^{n_{ij}} h_{ijk}(\beta) = 1.$$

For cells with $g(i) = g$, $h_{ijk}(\beta) = \mu_{ijk}(\beta)/S_g(\beta)$. Summing over all such cells and using $S_g(\beta) := \sum_{i:g(i)=g} \sum_{j,k} \mu_{ijk}(\beta)$ by definition:

$$\sum_{i:g(i)=g} \sum_{j,k} h_{ijk}(\beta) = \frac{\sum_{i:g(i)=g} \sum_{j,k} \mu_{ijk}(\beta)}{S_g(\beta)} = \frac{S_g(\beta)}{S_g(\beta)} = 1.$$

Remark 9 *This identity holds for every β , not merely at the converged MLE. General canonical-link GLMs only guarantee $\text{tr}(H(\hat{\beta})) = p$ at convergence, with no per-group decomposition available before fitting. Because $\mathcal{I}(\beta)$ is exactly block-diagonal here — a direct consequence of X having exactly one nonzero entry per row — the leverage sum splits identically as one unit per group at every β . This is what collapses the penalty term in (15) to a constant $1/2$ per group before the score equation is even solved, giving a closed form rather than requiring IRLS.*

Proposition 13 (Relation to CR2 leverage share) *For patient i with $n_i = 1$ image, $\sum_{k=1}^{N_i} h_{i1k}(\hat{\beta}) = \ell_i$, the CR2 leverage share of Definition 3, evaluated at $\hat{\beta}$.*

$\sum_k h_{i1k}(\hat{\beta}) = \sum_k \hat{\mu}_{i1k}/S_{g(i)} = T_i/S_{g(i)} = \ell_i$, using $T_i = \sum_k \hat{\mu}_{i1k}$ over patient i 's cells. Summing (15) over cells in group g and applying Proposition 12:

$$U_g^*(\beta) = Y_g - S_g(\beta) + \frac{1}{2}.$$

Setting $U_g^*(\hat{\beta}_g^{\text{Firth}}) = 0$ and substituting $S_g(\beta) = e^{\beta_g} D_g$:

$$\boxed{\hat{\beta}_g^{\text{Firth}} = \log\left(\frac{Y_g + 0.5}{D_g}\right)}. \quad (16)$$

No IRLS loop, convergence check, or fitting call is required: the Poisson case reduces to two scalar sums already available from the raw cell counts and density offsets.

10.4 Comparison to the Crude Per-Cell Correction

A common ad hoc rescue for the boundary problem of Section 10.1 is to add a constant $c = 0.5$ to every cell’s response before fitting the ordinary MLE. Writing $N_g := \sum_{i:g(i)=g} N_i$ for the total number of REF cells in group g , this shifts the group total by $0.5N_g$ rather than 0.5:

$$\hat{\beta}_g^{\text{crude}} = \log\left(\frac{Y_g + 0.5N_g}{D_g}\right).$$

Remark 10 (The N_g -fold gap) *Comparing to (16), the crude fix adds N_g times as many pseudo-counts to the group total as Firth does, with N_g routinely in the thousands for `spicyGLM`’s design. The value $c = 0.5$ is conventionally motivated by the Haldane–Anscombe correction for a binomial 2×2 table, where each cell contributes one unit of exposure; it carries no such justification for an offset-Poisson design where many cells share exposure through a common group parameter. Firth’s correction is derived directly from the information geometry of this design (Proposition 12) and adds exactly 0.5 to the group total regardless of N_g .*

Remark 11 (Scope) *This exactness relies on `spicyGLM`’s specific no-intercept, one-parameter-per-group, rank-1-per-patient design (Section 6.1). Introducing a covariate shared across groups would break the block-diagonal structure of $\mathcal{I}(\beta)$, and Proposition 12’s per-group sum-to-one identity would no longer hold at arbitrary β ; the penalty gradient would then require genuine IRLS at each step, as in general Firth-corrected GLMs (e.g. `brglm2`).*

11 Diagnostics

11.1 Motivation

The point estimate $\hat{\beta}$ and the CR2 sandwich estimator $\hat{V}_{\text{CR2}}$ (Section 4) together determine the reported effect size and its associated test of significance. Neither object, on its own, indicates whether that result actually reflects a population-level pattern shared broadly across patients, or whether it is substantially attributable to one patient, or one image within a patient. This section develops a set of diagnostic quantities, derived directly from objects already defined in Sections 4–6, that address three distinct questions:

1. Could a given patient or image structurally dominate its group, based on cell counts and target density alone, without reference to the observed neighbour counts and independent of the fitted model? Section 11.2 (*leverage*) answers this from the design alone.
2. Did a given patient or image’s observed neighbour counts disproportionately drive the reported variance estimate $\hat{V}_{\text{CR2}}$, and hence the p-value? Section 11.3 (*influence*) answers this from the fitted residuals.
3. Did a given patient’s observed neighbour counts move the reported effect size $\hat{\beta}$ itself? Section 11.4 (*point-estimate shift*) answers this directly, independent of the variance machinery entirely.

These three quantities are complementary: each is constructed from a different part of the model, and a patient or image may score highly on one pair while being unremarkable on the other pairs. Section 11.5 discusses how the three are read together. Section 11.6 addresses their aggregation across the many cell-type-pair comparisons a typical spicyGLM analysis produces.

11.2 Leverage

11.2.1 Patient-Level Leverage

Definition 3 (Leverage share) *For patient i , the leverage share is*

$$\ell_i := \frac{T_i}{S_{g(i)}}, \quad T_i = \sum_{j=1}^{n_i} n_{ij} \hat{\mu}_{ij}, \quad S_g = \sum_{i:g(i)=g} T_i,$$

with T_i and S_g as defined in Section 6.3.

ℓ_i is patient i 's share of their group's total weighted mass $S_{g(i)}$, and is computable directly from the raw per-image cell counts and density offsets. In the one-image-per-patient case, $\ell_i = N_i \alpha_i$, where α_i is the leverage scalar of Section 5.3.

Proposition 14 $0 < \ell_i < 1$, with $\ell_i \rightarrow 1$ if and only if patient i is the sole patient contributing to group $g(i)$.

Immediate from Proposition 3 in the one-image case (since $\ell_i = N_i \alpha_i < 1$); the general case follows identically, as $S_{g(i)} = \sum_{i':g(i')=g(i)} T_{i'} \geq T_i$, with equality iff patient i is the sole contributing patient in the group.

The boundary $\ell_i \rightarrow 1$ corresponds exactly to the group-singularity condition of Remark 3: CR2 is undefined when a group contains fewer than two patients with usable data, and ℓ_i provides a continuous measure of proximity to that boundary.

11.2.2 Image-Level Decomposition

Definition 4 (Within-patient leverage share) *For image j of patient i ,*

$$\ell_{ij} := \frac{n_{ij} \hat{\mu}_{ij}}{T_i}.$$

By construction, $\sum_{j=1}^{n_i} \ell_{ij} = 1$, and the two quantities compose exactly:

$$\ell_i \cdot \ell_{ij} = \frac{n_{ij} \hat{\mu}_{ij}}{S_{g(i)}},$$

image j 's leverage share of the entire group, directly comparable across images belonging to different patients within the same pair.

11.3 Influence

11.3.1 Motivation

Leverage (Section 11.2) is a property of the design alone and does not reference the fitted residuals $\hat{r}_i$; it measures whether a patient or image *could* dominate a group, not whether their data actually did anything unusual relative to the fitted model. A patient with high leverage whose data closely matches the fitted rate contributes little to the realised variance estimate, while a patient with low leverage can still contribute substantially if their residual is large. *Influence* is constructed to capture this second, outcome data-dependent notion directly.

11.3.2 Derivation

Let $L = (-1, 1)$ denote the contrast vector for $\beta_2 - \beta_1$ (Section ??). Since $\mathcal{B} = \text{diag}(1/S_1, 1/S_2)$,

$$LB = \left(-\frac{1}{S_1}, \frac{1}{S_2} \right) =: w^\top.$$

Because $X_i = \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top$ has rank 1, for any N_i -vector u ,

$$X_i^\top u = \mathbf{e}_{g(i)} (\mathbf{1}_{N_i}^\top u).$$

Define $s_i := \mathbf{1}_{N_i}^\top A_i \hat{r}_i$. Then $X_i^\top A_i \hat{r}_i = s_i \mathbf{e}_{g(i)}$, and each term of $\widetilde{\mathcal{M}}$ collapses to

$$X_i^\top A_i \hat{r}_i \hat{r}_i^\top A_i^\top X_i = s_i^2 \mathbf{e}_{g(i)} \mathbf{e}_{g(i)}^\top,$$

so that $\widetilde{\mathcal{M}}$ is exactly diagonal, with no cross-patient terms:

$$\widetilde{\mathcal{M}} = \begin{pmatrix} \sum_{i:g(i)=1} s_i^2 & 0 \\ 0 & \sum_{i:g(i)=2} s_i^2 \end{pmatrix}.$$

Defining $e_i := w_{g(i)} s_i$, it follows that

$$L \hat{V}_{\text{CR2}} L^\top = w^\top \widetilde{\mathcal{M}} w = \sum_{i=1}^m e_i^2 =: \hat{v}_L. \quad (17)$$

11.3.3 Definition

Definition 5 (Patient-level influence) For patient i ,

$$\text{Infl}_i := \frac{e_i^2}{\hat{v}_L}.$$

Proposition 15 $\sum_{i=1}^m \text{Infl}_i = 1$.

Immediate from (17): $\sum_i \text{Infl}_i = \frac{1}{\hat{v}_L} \sum_i e_i^2 = \hat{v}_L / \hat{v}_L = 1$.

Infl_i is the exact fractional contribution of patient i to the CR2 variance estimate that determines the reported Wald statistic and p-value for a given pair.

11.3.4 Relationship to Leave-One-Out Influence

Infl_i admits a narrower definition compared to the more rigorous classical leave-one-out (e.g. Cook's-distance-style) diagnostic. Writing $\hat{v}_L^{(-i)} := \sum_{k \neq i} e_k^2$ for $\hat{v}_L$ with patient i 's term removed,

$$\hat{v}_L - \hat{v}_L^{(-i)} = e_i^2, \quad \text{so} \quad \text{Infl}_i = \frac{\hat{v}_L - \hat{v}_L^{(-i)}}{\hat{v}_L},$$

the exact fractional decrease in $\hat{v}_L$ from removing patient i 's term. This is distinct from the fractional decrease that would result from genuinely deleting patient i and refitting: such a refit would also alter $S_{g(i)}$, and hence A_k and e_k for every other patient k in group $g(i)$, and $\hat{\beta}_{g(i)}$ itself. $\hat{v}_L^{(-i)}$ holds these fixed.

11.3.5 Image-Level Decomposition

Partitioning s_i by image, since images partition a patient's cells exactly,

$$s_{ij} := \sum_{k \in \text{image } j} (A_i \hat{r}_i)_k = r_{ij}^{\text{sum}} + \sqrt{n_{ij}} (\bar{A}_i d)_j, \quad e_{ij} := w_{g(i)} s_{ij},$$

where $r_{ij}^{\text{sum}} = \sum_{k \in \text{image } j} \hat{r}_{ik}$ and $d = P_i^\top \hat{r}_i$ (Section 6.9). By construction $e_i = \sum_j e_{ij}$ exactly. Because $e_i^2 = \left(\sum_j e_{ij}\right)^2 \neq \sum_j e_{ij}^2$ in general (the cross terms between images of the same patient do not vanish), a share built directly from e_{ij}^2 does not partition Infl_i exactly. Instead:

Definition 6 (Image-level influence) *For image j of patient i ,*

$$\text{Infl}_{ij} := \text{Infl}_i \cdot \frac{e_{ij}}{e_i} = \frac{e_{ij} e_i}{\hat{v}_L}.$$

Proposition 16 $\sum_{j=1}^{n_i} \text{Infl}_{ij} = \text{Infl}_i$ exactly, and consequently $\sum_i \sum_j \text{Infl}_{ij} = 1$.

$\sum_j \text{Infl}_{ij} = \text{Infl}_i \cdot \frac{1}{e_i} \sum_j e_{ij} = \text{Infl}_i \cdot \frac{e_i}{e_i} = \text{Infl}_i$; summing over i and applying Proposition 15 gives the second claim.

Unlike Infl_i , Infl_{ij} is signed: $\text{Infl}_{ij} < 0$ indicates that image j 's corrected residual contribution e_{ij} opposes the direction of its patient's net contribution e_i . Because Infl_{ij} sums to 1 across every image of every patient in a pair, image-level influence values are only directly comparable across the whole pair, not merely within a single patient.

11.4 Point-Estimate Shift

Leverage and influence are constructed entirely from the variance side of the Wald statistic $t = L\hat{\beta}/\sqrt{\hat{v}_L}$; neither addresses whether $\hat{\beta}$ itself — the reported log rate ratio — is substantially attributable to one patient's data. Because $\hat{\beta}_g$ is the closed-form Firth estimate

$$\hat{\beta}_g = \log\left(\frac{Y_g + 0.5}{D_g}\right), \quad Y_g = \sum_{i:g(i)=g} y_i, \quad D_g = \sum_{i:g(i)=g} d_i,$$

where y_i, d_i are patient i 's totals, deleting a single patient's contribution and recomputing is exact and requires no refit:

$$\hat{\beta}_{g(i)}^{(-i)} := \log\left(\frac{Y_{g(i)} - y_i + 0.5}{D_{g(i)} - d_i}\right).$$

Definition 7 (Point-estimate shift) *For patient i ,*

$$\Delta_i := \hat{\beta}_{g(i)} - \hat{\beta}_{g(i)}^{(-i)}.$$

Δ_i is the exact change in the reported log rate ratio that would result from removing patient i 's data entirely, holding every other patient fixed. Unlike ℓ_i and Infl_i , Δ_i is not defined at the image level: β_g is a group-level parameter with no image-level (or, more precisely, no sub-patient) grain in the mean model of Section 6 — $g(i)$ is assigned per patient, and no image-specific term exists for Δ_i to be attributed to below the patient. Patient-level deletion is therefore the finest level at which this diagnostic is well-posed.

11.5 Reading the Diagnostics Together

Returning to the three questions of Section 11.1: ℓ_i (Section 11.2) answers whether a patient or image *could* structurally dominate its group; Infl_i (Section 11.3) answers whether it *did* dominate the reported variance estimate; and Δ_i (Section 11.4) answers whether it moved the reported effect size directly. These three need not agree: a patient with high ℓ_i and low Infl_i has data consistent with the fitted rate despite occupying a structurally dominant position, while a patient with low ℓ_i and high Infl_i is a minor contributor by design whose data was nonetheless unexpected. A pair's reported p-value is additionally governed by the degrees of freedom ν (Section ??), which — being a function of leverage-type quantities alone, with no dependence on realised residuals — can be small even when no individual patient is flagged by ℓ_i or Infl_i ; a full reading of a given pair's result should account for ν alongside the three patient-level diagnostics above, rather than in place of them.

11.6 Cross-Pair Aggregation

11.6.1 The Comparability Problem

For a fixed pair, ℓ_i and Infl_i are normalised across the patients contributing to that pair's fit, summing to 1 over those patients (or, for Infl_{ij} , over their images), and this normalisation is recomputed independently for every pair, since each pair has its own fit and its own group totals S_1, S_2 . The number of contributing patients m varies from pair to pair since there are patients whose data does not include both cell types in a given comparison. Hence, since patients and their images are excluded from that pair's fit entirely, m can differ substantially even within a single dataset for different pairs.

Since ℓ_i and Infl_i each sum to 1 across a pair's m contributing patients, the value either would take under a perfectly even split of leverage or influence. In other words if no patient contributed disproportionately more than any other, their baseline value should be $1/m$. This is the level at which a patient's raw ℓ_i or Infl_i indicates neither above- nor below-average standing within their group. Because the baseline itself depends on m , raw values of

ℓ_i or Infl_i are not directly comparable across pairs of different sizes: an unremarkable patient in a 60-patient group and an unremarkable patient in a 5-patient group carry baseline values of $1/60$ and $1/5$ respectively, so identical raw values can represent very different degrees of relative dominance.

11.6.2 Relative Leverage, Influence, and Point-Estimate Shift

Definition 8 (Relative leverage / influence) *For patient i in a pair with m contributing patients in group $g(i)$,*

$$\tilde{\ell}_i := \ell_i \cdot m, \quad \widetilde{\text{Infl}}_i := \text{Infl}_i \cdot m,$$

with the analogous image-level definitions

$$\tilde{\ell}_{ij} := \ell_{ij} \cdot n_i, \quad \widetilde{\text{Infl}}_{ij} := \text{Infl}_{ij} \cdot N_{g(i)}^{\text{img}},$$

where n_i is the number of images belonging to patient i (the population ℓ_{ij} is normalised over, Definition 4), and $N_{g(i)}^{\text{img}} := \sum_{i': g(i')=g(i)} n_{i'}$ is the total number of images across every patient in group $g(i)$ (the population Infl_{ij} is normalised over, Proposition 16). The two image-level diagnostics use different baselines because ℓ_{ij} sums to 1 within a single patient's own images, while Infl_{ij} sums to 1 across every image in the entire group.

A value of $\tilde{\ell}_i = 1$ (respectively $\widetilde{\text{Infl}}_i = 1$) corresponds exactly to the share a patient would receive under a perfectly even split of leverage (resp. influence) across their group; values above 1 indicate above-average, and below 1 below-average, standing — on a scale that no longer depends on m , and is therefore comparable across pairs of different sizes.

Point-estimate shift has no such baseline. Δ_i is not a share of a pair-wide or group-wide total in the sense ℓ_i and Infl_i are — there is no even-split value Δ_i would take if no patient contributed disproportionately, since Δ_i measures an absolute shift in a log-scale quantity rather than a proportion of a fixed pool. No relative rescaling $\widetilde{\Delta}_i$ is therefore defined; $|\Delta_i|$ is carried forward to the percentile-normalisation step of Section 11.6.3 unrescaled.

11.6.3 Within-Pair Percentile Normalisation

Relative diagnostics correct for differences in group *size* across pairs, but not for differences in group *shape*: two pairs with the same m can have very different distributions of ℓ_i or Infl_i across their patients (near-uniform in one, sharply peaked in another), so that equal values of $\tilde{\ell}_i$ or $\widetilde{\text{Infl}}_i$ need not represent equally extreme standing within their respective pairs. To obtain a quantity comparable across pairs regardless of shape, each relative diagnostic value is converted to its percentile rank within its own pair's distribution before any cross-pair comparison is made — e.g. for $\widetilde{\text{Infl}}_{ij}$, its rank among every $\widetilde{\text{Infl}}_{i'j'}$ value in the same pair, expressed as a proportion in $(0, 1]$. Since Δ_i carries no relative rescaling (Section 11.6.2), its percentile rank is computed directly on $|\Delta_i|$, ranked among every patient contributing to the pair — the same pair-wide population used for $\widetilde{\text{Infl}}_i$, since Δ_i , like Infl_i , is defined relative to a single group's pooled estimate rather than scoped within one condition group only, in contrast to ℓ_i . Because β_g has no image-level grain (Section 11.4), no image-level percentile rank is defined for Δ_i .

11.6.4 Cross-Pair Flagging

Fixing a percentile threshold (e.g. the top 5% arbitrarily) applied uniformly to the within-pair percentile ranks, a patient is flagged in a given pair if its rank exceeds the threshold. For a patient appearing in n_{pairs} pairs, flagged in n_{flagged} of them, the raw proportion $n_{\text{flagged}}/n_{\text{pairs}}$ is unstable when n_{pairs} is small. A Wilson score interval is computed around this proportion, and patients are ranked by the interval's lower bound rather than the raw proportion, so that a high flagging rate based on few pair-appearances is appropriately discounted relative to an equally high rate supported by many.

11.6.5 Image-Level Mean-Share Summaries

Cross-pair flagging establishes, at the patient level, which patients are repeatedly extreme across many cell-type pairs. Given such a patient, the natural next question is which of their images is responsible: does the patient's cross-pair signature arise from one image consistently, or is it spread across several? Answering this requires a decomposition of the patient-level quantity down to image level, averaged across the same cell-type pairs used for flagging.

Summary statistics. Four averages serve this purpose: the *within-patient leverage mean share*, the *whole-group leverage mean share*, the *within-patient influence mean share*, and the *whole-pair influence mean share*, each reusing a share pair already established in Sections 11.2–11.3.

Definition 9 (Mean image-level shares) For image j of patient i , averaged over the set P_{ij} of cell-type pairs in which image j has usable data,

$$\begin{aligned}\bar{\ell}_{ij} &:= \frac{1}{|P_{ij}|} \sum_{p \in P_{ij}} \ell_{ij,p} && (\text{within-patient leverage mean share}), \\ \overline{\ell_i \ell_{ij}} &:= \frac{1}{|P_{ij}|} \sum_{p \in P_{ij}} \ell_{i,p} \ell_{ij,p} && (\text{whole-group leverage mean share}), \\ \overline{e_{ij}/e_i} &:= \frac{1}{|P_{ij}|} \sum_{p \in P_{ij}} \frac{e_{ij,p}}{e_{i,p}} && (\text{within-patient influence mean share}), \\ \overline{\text{Infl}_{ij}} &:= \frac{1}{|P_{ij}|} \sum_{p \in P_{ij}} \text{Infl}_{ij,p} && (\text{whole-pair influence mean share}).\end{aligned}$$

Interpretation. $\bar{\ell}_{ij}$ and $\overline{\ell_i \ell_{ij}}$, built from leverage shares, lie in $(0, 1)$ throughout, exactly as ℓ_{ij} and $\ell_i \ell_{ij}$ do at the single-pair level (Section 11.2): a value near $1/n_i$ (respectively near the group's even-split baseline) indicates the image is an unremarkable, evenly-weighted contributor; a value substantially above indicates it structurally dominates its patient's (or its group's) leverage on average.

$\overline{e_{ij}/e_i}$ and $\overline{\text{Infl}_{ij}}$ require more care, since e_{ij} and Infl_{ij} are signed (Section 11.3) and unbounded, even within a single pair.

Reading e_{ij}/e_i within a single pair. e_{ij} is image j 's own corrected deviation from the fitted model's prediction for that cell-type pair; $e_i = \sum_j e_{ij}$ is the same quantity summed across the patient's images. The ratio compares only these two numbers to each other:

- **Direction** is the sign: whether image j 's observed neighbour counts deviated from the fitted prediction in the same sense as the patient's combined deviation, or the opposite sense.
- **Magnitude** is $|e_{ij}/e_i|$: how large image j 's own deviation is relative to the patient's combined deviation.

Four cases follow directly from $e_i = \sum_j e_{ij}$, describing how image j 's own deviation from the fitted model compares to the deviation summed across all of patient i 's images:

- $e_{ij}/e_i = 1$. Image j 's deviation matches the patient's total exactly, in both direction and size. This is the only possible value when patient i has a single image ($n_i = 1$): with nothing else to sum, $e_{ij} = e_i$ trivially, and the ratio equals 1 in every pair the patient appears in.
- $0 < e_{ij}/e_i < 1$. Same direction as the total, smaller in size: image j accounts for part, but not all, of the patient's overall deviation.
- $e_{ij}/e_i > 1$. Same direction as the total, larger in size: image j 's own deviation exceeds the patient's combined total.
- $e_{ij}/e_i < 0$. Opposite direction from the total: image j deviated from the fitted prediction in the sense opposite to the patient's combined result.

None of these four cases makes any claim about what any other image of the same patient contributed: the ratio describes only image j 's relationship to the patient's already-combined total, not the individual behaviour of any other image making up that total.

Reading the cross-pair mean $\overline{e_{ij}/e_i}$. Each term in this average is computed from a different cell-type pair, with its own fitted model and its own e_i . The mean therefore reports image j 's *typical* direction and magnitude relative to its patient's total, across the pairs it appears in — it does not imply that the same one of the four cases above holds in every individual pair. A mean such as 1.4 could arise from pairs that were individually unremarkable (ratios near 0.5) mixed with pairs where the ratio was substantially larger; the average cannot distinguish these. To establish which specific pairs drive an unusual mean, the per-pair values in Table 3 must be inspected directly; the mean itself carries no information about any individual pair once computed.

$\overline{\text{Infl}}_{ij}$: magnitude rescaled, sign unchanged. Since $\text{Infl}_{ij} = \text{Infl}_i \times (e_{ij}/e_i)$ and $\text{Infl}_i \geq 0$ always, multiplying by Infl_i never changes sign: Infl_{ij} is negative exactly when e_{ij}/e_i is, and its sign is read directly, exactly as for e_{ij}/e_i above. Since Infl_i itself has no sign, it plays no role in the direction reading — only in magnitude. The four cases become:

- $\text{Infl}_{ij} = \text{Infl}_i$: same magnitude as the patient’s total (equivalent to $e_{ij}/e_i = 1$).
- $0 < \text{Infl}_{ij} < \text{Infl}_i$: smaller magnitude, positive.
- $\text{Infl}_{ij} > \text{Infl}_i$: larger magnitude, positive.
- $\text{Infl}_{ij} < 0$: negative.

The magnitude comparison is against Infl_i , not the literal number 1; the direction is read directly from Infl_{ij} ’s own sign, not from any comparison to Infl_i , which carries no direction of its own.

Remark 12 (The mean does not preserve the exact-sum property) *Within a single pair p , $\sum_j \ell_{ij,p} = 1$ and $\sum_j e_{ij,p}/e_{i,p} = 1$ exactly (Definition 4, Section 11.3). This does not carry over to Definition 9 whenever a patient’s images have unequal pair coverage — i.e. $P_{ij} \neq P_{ij'}$ for two images $j \neq j'$ of the same patient, which occurs whenever one image lacks usable co-occurring data in a pair another image of the same patient does have. Each image’s mean is then computed over a different underlying set of pairs, and $\sum_j e_{ij}/e_i$ need not equal 1.*

12 Extension to the Binomial GLM (Fixed- k Nearest-Neighbour Design)

12.1 Motivation and Model Setup

Section 3.1’s Poisson model defines a REF cell’s neighbourhood via a fixed physical radius r , so the number of cells at risk of being TARGET, N_i , varies with local tissue density — a cell in a dense region has many more candidate neighbours than one in a sparse region. An alternative neighbourhood definition fixes the *number* of candidate neighbours instead: for each REF cell, search its k nearest neighbours (by Euclidean distance, irrespective of cell type), and ask what fraction of those k are TARGET type. Because the denominator k is now fixed by design rather than varying with density, the response is naturally a proportion out of a known total, and Binomial is the corresponding exponential-family model, in the same way Poisson was the natural choice for an open-ended count against a variable exposure.

Model. Let k denote the fixed number of nearest neighbours searched per REF cell. For REF cell ℓ in image j of patient i , let $Y_{ij\ell} \in \{0, \dots, k\}$ denote the number of that cell’s k nearest neighbours that are TARGET type. Let

$$p_{0,ij} = \frac{|\{\text{TARGET cells in image } j \text{ of patient } i\}|}{|\{\text{all cells in image } j \text{ of patient } i\}|}$$

denote the image-level background proportion of TARGET cells — the Binomial analogue of Poisson’s $\widehat{\text{den}}_{ij}$, playing the same role (an image-level null-expectation quantity derived from TARGET abundance) but expressed as a proportion rather than an intensity. Write $o_{ij} := \text{logit}(p_{0,ij})$.

The model is a no-intercept Binomial GLM with logit link, with the same two-group, one-parameter-per-group structure as Section 3.1:

$$Y_{ij\ell} \sim \text{Binomial}(k, \pi_{ij\ell}), \quad \text{logit}(\pi_{ij\ell}) = \beta_{g(i)} + o_{ij}.$$

The covariate vector is unchanged from Poisson: $x_{ij\ell} = \mathbf{e}_{g(i)}$.

Interpretation of β . Rearranging the link equation,

$$\frac{\pi_{ij\ell}/(1 - \pi_{ij\ell})}{p_{0,ij}/(1 - p_{0,ij})} = e^{\beta_{g(i)}},$$

so e^{β_g} is an **odds ratio**: the odds that a REF cell's k -nearest-neighbour set contains a TARGET cell, relative to the odds implied by that image's own background TARGET prevalence.

$$e^{\beta_g} > 1 \implies \text{enrichment}, \quad e^{\beta_g} < 1 \implies \text{depletion}.$$

Remark 13 (Odds ratio, not rate ratio) *This is a different quantity from Poisson's e^{β_g} . Poisson's coefficient is a ratio of expected counts (a rate ratio), while Binomial's is a ratio of odds of a compositional event. The primary contrast $\beta_1 - \beta_2$ should accordingly be reported and discussed as a log odds-ratio contrast throughout, not translated into rate-ratio language.*

12.2 Notation Changes

This section states only the changes from Table 2: symbols that are new, and symbols whose definitions change. Every symbol not listed here (indexing objects $m, n_i, n_{ij}, N_i, N, g(i)$; the design matrix X_i ; the sandwich/CR2 symbols $\mathcal{I}, \mathcal{B}, H_X, Q_i, G_i, A_i, \mathcal{M}, \widetilde{\mathcal{M}}, \hat{V}_{\text{CR2}}$) is carried over unchanged in both symbol and meaning.

New symbols.

Symbol	Dimension	Description
k	scalar	number of nearest neighbours searched per REF cell (fixed, global)
$Y_{ij\ell}$	scalar	count of TARGET cells among cell ℓ 's k nearest neighbours
$p_{0,ij}$	scalar	image-level background TARGET proportion
$o_{ij} = \text{logit}(p_{0,ij})$	scalar	logit-scale offset for image j of patient i
$\pi_{ij\ell}$	scalar	modelled probability that a given one of cell ℓ 's k neighbours is TARGET

Redefined symbols.

Symbol	Poisson (Table 2)	Binomial (this section)
μ_i (fitted mean vector)	$\hat{\mu}_{ijk} = \exp(\hat{\beta}_{g(i)}) \widehat{\text{den}}_{ij}$	$\hat{\mu}_{ij\ell} = k\hat{\pi}_{ij\ell}$
V_i (working variance)	$V_i = \text{diag}(\mu_i)$	$V_i = \text{diag}(k\pi_i(1 - \pi_i))$

Remark 14 (Mean and variance no longer coincide) *For the Poisson GLM, $V_i = \text{diag}(\mu_i)$ identically — the working variance matrix and the fitted mean vector are the same object, so no notational distinction between “ $\hat{\mu}$ as fitted value” and “ $\hat{\mu}$ as the GLM variance function” was necessary anywhere in Sections 4–9. For the Binomial GLM, $\mu_i = k\pi_i$ and $V_i = k\pi_i(1 - \pi_i)$ are numerically distinct functions of π_i . Consequently, every place Sections 4–9 write $\hat{\mu}_{ij}$ or $\hat{\mu}_i$ must be read according to the role that quantity plays: occurrences used to compute fitted values or residuals ($\hat{r}_i = Y_i - \hat{\mu}_i$) require the true mean $k\pi_i$; occurrences feeding $\Lambda_i, \mathbf{f}_i, T_i, S_g$, and everything built from them (the entirety of the CR2 sandwich and Satterthwaite degrees-of-freedom machinery) require the working variance $k\pi_i(1 - \pi_i)$. We write $\mathbf{v}_i \in R^{N_i}$ for this working-variance vector, constant on each image block (since $\pi_{ij\ell}$ does not depend on ℓ) exactly as $\boldsymbol{\mu}_i$ was in Section 6.1, so that $V_i = \text{diag}(\mathbf{v}_i)$. Section 12.3 states this correspondence once, as a single substitution rule ($\boldsymbol{\mu}_i \mapsto \mathbf{v}_i$), rather than re-deriving each downstream result.*

12.3 The CR2 Adjustment and Degrees of Freedom Survive Unchanged

Sections 4–6 derive the sandwich variance, the BRL criterion, the simplification $Q_i = (I - H_{X,ii})V_i$, the DPR1 structure of G_i , the closed-form within-image eigenpairs, the reduced eigenvalue problem, and the assembly of A_i (Eq. 11), together with its complexity analysis. Section 9 additionally derives the Satterthwaite degrees of freedom from these same objects. Every step in this entire chain uses only two structural facts: V_i is diagonal, and $X_i = \mathbf{1}_{N_i} \mathbf{e}_{g(i)}^\top$ has rank one with all rows equal. Neither fact is specific to the Poisson family — the Binomial GLM of Section 12.1 satisfies both, with $V_i = \text{diag}(\mathbf{v}_i)$, $\mathbf{v}_i$ as in Remark 14.

Proposition 17 (Substitution principle) *Every proposition, lemma, and closed-form result in Sections 4–9 holds for the Binomial GLM of Section 12.1 under the substitution $\boldsymbol{\mu}_i \mapsto \mathbf{v}_i$ (equivalently $\hat{\mu}_{ij} \mapsto \hat{V}_{ij} := k\hat{\pi}_{ij}(1 - \hat{\pi}_{ij})$) applied wherever $\hat{\mu}_{ij}$ appears in the construction of $\Lambda_i, \mathbf{f}_i, T_i, S_g, H_{X,ii}, G_i, A_i, \tilde{\tau}_j, h_j, P_j^{\text{diag}}$. No other change to any proof is required.*

[Justification] For Sections 4–6: $Q_i = (I - H_{X,ii})V_i$ (Proposition 2) is proved by expanding $Q_i = \sum_j (I - H_X)_{ij} V_j (I - H_X)_{ij}^\top$ and collapsing the cross-patient sum via $\mathcal{BI} = I$; this uses $\mathcal{I} = \sum_j X_j^\top V_j X_j$ generically and never the specific functional form of V_j . Section 6.3’s derivation of $T_i = \sum_j n_{ij} \hat{\mu}_{ij}$, $H_{X,ii} = \boldsymbol{\mu}_i \mathbf{1}_{N_i}^\top / S_{g(i)}$, and $G_i = \Lambda_i - \frac{1}{S_{g(i)}} \mathbf{f}_i \mathbf{f}_i^\top$ uses only V_i diagonal and X_i rank-1: $T_i = \mathbf{1}_{N_i}^\top V_i \mathbf{1}_{N_i}$ is a sum over the diagonal of V_i by definition,

whatever that diagonal contains; $\Lambda_i = D_i V_i D_i = \text{diag}(\boldsymbol{\mu}_i)^2$ and $\mathbf{f}_i = D_i \boldsymbol{\mu}_i = \boldsymbol{\mu}_i^{3/2}$ follow purely from $D_i = V_i^{1/2}$ acting elementwise, again independent of what V_i 's diagonal entries represent. Sections 6.2–6.8 (the image-indicator subspace P_i, R_i , Proposition 4's closed-form eigenpairs, Proposition 5's invariance of $\text{range}(P_i)$, the reduced matrix $\bar{G}_i$, and the assembly of A_i in Eq. 11) use only that Λ_i and $\mathbf{f}_i$ are constant on each image block, which holds for any block-constant V_i regardless of its functional origin.

For Section 9, the same holds by identical inspection. Lemma 3 ($\partial\hat{\beta}/\partial Y_k = \mathcal{B}X_k^\top$) follows from differentiating the score equation and uses only $\partial\mu_l/\partial\beta = V_l X_l$, true for any canonical-link GLM. Lemma 4's loading vector g_j is built from A_j , $\tilde{\tau}_j = \mathbf{1}_{N_j}^\top A_j \hat{\mu}_j$, and $S_{g(j)}$ — the first inherited directly from Proposition 17's CR2 result, the latter two sums over diagonal entries of V_i -derived quantities, agnostic to family. Proposition 9 and the Gram-matrix reduction (Proposition 10) are pure linear algebra on $\Sigma = \text{blockdiag}(V_1, \dots, V_m)$, again using only block-diagonal structure. Theorem 9.5's closed form for P_{jk} is derived entirely from $[g_j]_l$'s definition and $\mathbf{1}_{N_j}^\top V_j \mathbf{1}_{N_j} = T_j$ -type identities, all inherited from the same V_i -diagonal argument. No step in Section 9 uses Poisson's mean-variance identity beyond $\hat{\mu}_i$ standing in for V_i 's diagonal, which Remark 14 replaces with $\mathbf{v}_i$.

Consequence — CR2. Restating the final boxed results with the substitution applied: *One image per patient* (Section 5): with $\hat{V}_i := k\hat{\pi}_i(1 - \hat{\pi}_i)$ and $\alpha_i := \hat{V}_i/S_{g(i)}$, $S_g := \sum_{i:g(i)=g} N_i \hat{V}_i$,

$$A_i = I_{N_i} + \frac{1}{N_i} \left(\frac{1}{\sqrt{1 - \alpha_i N_i}} - 1 \right) \mathbf{1}_{N_i} \mathbf{1}_{N_i}^\top,$$

identical in form to Eq. 8.

Multiple images per patient (Section 6): with $\bar{\Lambda}_i = \text{diag}(\hat{V}_{i1}^2, \dots, \hat{V}_{i,n_i}^2)$, $(\bar{\mathbf{f}}_i)_j = \sqrt{n_{ij}} \hat{V}_{ij}^{3/2}$, $\bar{D}_i = \text{diag}(\hat{V}_{i1}^{1/2}, \dots)$, and $\bar{V}_i, \bar{\Lambda}_i^{\text{eig}}$ from `eigen()` applied to $\bar{G}_i = \bar{\Lambda}_i - \frac{1}{S_{g(i)}} \bar{\mathbf{f}}_i \bar{\mathbf{f}}_i^\top$ (Section 6.12; `dpr1eig` is not used for either family in this implementation),

$$A_i = I_{N_i} + P_i \left(\bar{D}_i \bar{V}_i (\bar{\Lambda}_i^{\text{eig}})^{-1/2} \bar{V}_i^\top \bar{D}_i - I_{n_i} \right) P_i^\top,$$

identical in form to Eq. 11.

Consequence — degrees of freedom. $\tilde{\tau}_j$ is defined by differentiating $e_j(\beta') = w_{g(j)} \mathbf{1}_{N_j}^\top A_j (Y_j - \mu_j(\beta'))$ with respect to β , which hits $\partial\mu_j/\partial\beta = V_j X_j$ (the general canonical-link identity of Lemma 3's proof) — *not* μ_j itself. Writing $\mathbf{v}_j := V_j \mathbf{1}_{N_j}$ for the working-variance vector,

$$\tilde{\tau}_j := \mathbf{1}_{N_j}^\top A_j \mathbf{v}_j,$$

so $\tilde{\tau}_j$ is subject to the same $\boldsymbol{\mu}_i \mapsto \mathbf{v}_i$ substitution as $\Lambda_i, \mathbf{f}_i, T_i, S_g$ (Proposition 17). With $h_j := \tilde{\tau}_j/S_{g(j)}^{3/2}$ and $P_j^{\text{diag}} := w_{g(j)}^2 (A_j \mathbf{1}_{N_j})^\top V_j (A_j \mathbf{1}_{N_j})$, Theorem 9.5 and Corollary 4 apply verbatim:

$$\hat{\nu} = \frac{\left(\sum_j (P_j^{\text{diag}} - h_j^2) \right)^2}{\sum_{j,k:g(j)=g(k)} (h_j h_k)^2}.$$

Remark 15 (What this does not include, and a caution about $\tilde{\tau}_j$) Only $\hat{r}_i = Y_i - \hat{\mu}_i$ ($\hat{\mu}_i = k\hat{\pi}_i$, the true fitted mean) is exempt from the substitution of Proposition 17. $\tilde{\tau}_j$ is not exempt, despite being written with a $\hat{\mu}$ -like symbol in some treatments: it is built from $\partial\mu_j/\partial\beta = V_j X_j$, a working-variance object, and must use $\mathbf{v}_j$, not $\hat{\mu}_j$. For the Poisson GLM this distinction is invisible — $V_j \mathbf{1}_{N_j} = \hat{\mu}_j$ there, since $V(\mu) = \mu$ — so the Poisson document’s derivation (Section 9) can substitute $\hat{\mu}_j$ for $V_j \mathbf{1}_{N_j}$ freely, and the two symbols become interchangeable in that one section. Under Binomial, where $V(\mu) = \mu(1 - \mu/k) \neq \mu$, this coincidence no longer holds, and using $\hat{\mu}_j$ in place of $\mathbf{v}_j$ silently corrupts $\tilde{\tau}_j$, and hence the Satterthwaite degrees of freedom (Theorem 9.5), while leaving the point estimate and its variance $\hat{v}_L$ unaffected — since neither of those depends on $\tilde{\tau}_j$ at all.

12.4 Firth’s Correction Does Not Admit a Closed Form

Unlike Section 12.3, this section is not a substitution exercise: the closed form of Section 10 genuinely fails to survive, and the reason is worth deriving explicitly rather than simply asserted.

Binomial’s information and leverage. For the model of Section 12.1, write $\mu_{ij\ell}(\beta_g) := k\pi_{ij\ell}(\beta_g) = k \expit(\beta_g + o_{ij})$ for the mean at a general (not necessarily fitted) β_g , and $V(\mu_{ij\ell}(\beta_g)) := k\pi_{ij\ell}(\beta_g)(1 - \pi_{ij\ell}(\beta_g))$ for the working variance, in place of Poisson’s $V(\mu_{ijk}) = \mu_{ijk}$. Two group-level totals must be kept distinct here, though Section 10.1’s Poisson notation uses a single symbol $S_g(\beta)$ for both:

$$M_g(\beta_g) := \sum_{i:g(i)=g} \sum_{j,\ell} \mu_{ij\ell}(\beta_g) \quad (\text{sum of means, the plain score’s expected total}),$$

$$\mathcal{I}_g(\beta_g) := \sum_{i:g(i)=g} \sum_{j,\ell} V(\mu_{ij\ell}(\beta_g)) = \sum_{i:g(i)=g} \sum_{j,\ell} k\pi_{ij\ell}(\beta_g)(1 - \pi_{ij\ell}(\beta_g)) \quad (\text{Fisher information}).$$

The plain (unpenalized) Binomial score is $U_g(\beta_g) = Y_g - M_g(\beta_g)$. $\mathcal{I}_g(\beta_g)$ is the quantity matching the S_g already used throughout Section 12.3, evaluated at general β rather than at $\hat{\beta}$. For Poisson, $V(\mu) = \mu$, so $\mathcal{I}_g(\beta) = M_g(\beta)$ identically — the two roles coincide, which is why Section 10.1 can use a single symbol $S_g(\beta)$ for both; under Binomial they diverge, and conflating them gives an incorrect score equation.

Correspondingly, Definition 2’s Firth leverage, $h_{ijk}(\beta) := \mu_{ijk}(\beta) x_{ijk}^\top \mathcal{I}(\beta)^{-1} x_{ijk}$, is itself written using the mean $\mu_{ijk}(\beta)$ rather than the variance function $V(\mu_{ijk}(\beta))$ that its own derivation (the log-determinant expansion, Eq. 14) actually produces; this substitution is silent and correct only because $V(\mu) = \mu$ for Poisson. Generally,

$$h_{ijk}(\beta) := V(\mu_{ijk}(\beta)) x_{ijk}^\top \mathcal{I}(\beta)^{-1} x_{ijk},$$

which for the Binomial GLM of Section 12.1 specializes, using $x_{ijk}^\top \mathcal{I}(\beta)^{-1} x_{ijk} = 1/\mathcal{I}_{g(i)}(\beta)$ for our rank-1-per-group design, to

$$h_{ij\ell}(\beta) = \frac{k\pi_{ij\ell}(\beta)(1 - \pi_{ij\ell}(\beta))}{\mathcal{I}_{g(i)}(\beta)} = \frac{V(\mu_{ij\ell}(\beta))}{\mathcal{I}_{g(i)}(\beta)}.$$

With h defined this way, $\sum_{i:g(i)=g} \sum_{j,\ell} h_{ij\ell}(\beta) = \mathcal{I}_g(\beta)/\mathcal{I}_g(\beta) = 1$ for every β , by the definition of $\mathcal{I}_g(\beta)$ as the sum of exactly these terms — true for any canonical-link GLM with this design, not a Poisson-specific cancellation.

Where the derivation actually breaks. Equation 14’s general penalty gradient is

$$\frac{\partial}{\partial \beta_g} \log \det \mathcal{I}(\beta) = \sum_{ijk} V'(\mu_{ijk}) h_{ijk}(\beta) (x_{ijk})_g,$$

where V' is the *derivative of the variance function with respect to the mean*. Poisson’s specialization (Section 10.2) used $V(\mu) = \mu \implies V'(\mu) \equiv 1$, a constant — this is precisely what let V' drop out of the sum entirely, leaving $\sum_{ijk} h_{ijk}(\beta) (x_{ijk})_g$, which Proposition 12 then evaluated in closed form. For Binomial, $V(\mu) = \mu(1 - \mu/k)$ (writing the variance as a function of the mean $\mu = k\pi$), so

$$V'(\mu) = 1 - \frac{2\mu}{k} = 1 - 2\pi,$$

which is **not constant** — it depends on $\pi_{ij\ell}(\beta)$, and hence on β , cell by cell. Substituting into Eq. 12, the penalized score for group g is

$$U_g^*(\beta_g) = (Y_g - M_g(\beta_g)) + \frac{1}{2} \sum_{i:g(i)=g} \sum_{j,\ell} (1 - 2\pi_{ij\ell}(\beta_g)) h_{ij\ell}(\beta_g),$$

using $M_g(\beta_g)$, not $\mathcal{I}_g(\beta_g)$, in the first term. The correction term is a β -dependent weighted average of $(1 - 2\pi_{ij\ell})$ terms, weighted by each cell’s own leverage share. Unlike Poisson’s case, this does not collapse to a constant: it retains genuine dependence on the fitted probabilities in every image, through both $\pi_{ij\ell}(\beta)$ itself and its interaction with $h_{ij\ell}(\beta)$.

Proposition 18 (No closed-form Firth root) *For the Binomial GLM of Section 12.1, the Firth-penalized score equation*

$$U_g^*(\beta_g) = (Y_g - M_g(\beta_g)) + \frac{1}{2} \sum_{i:g(i)=g} \sum_{j,\ell} (1 - 2\pi_{ij\ell}(\beta_g)) h_{ij\ell}(\beta_g) = 0$$

is transcendental in β_g and admits no closed-form solution in general.

[Justification] $M_g(\beta_g) = \sum k \text{expit}(\beta_g + o_{ij})$, and the correction term through $h_{ij\ell}(\beta_g)$, both involve $\pi_{ij\ell}(\beta_g) = \text{expit}(\beta_g + o_{ij})$ evaluated at distinct offsets o_{ij} across images j (Section 12.1). Unlike Poisson, where $\mu_{ijk}(\beta_g) = e^{\beta_g} \widehat{\text{den}}_{ij}$ is linear in the single unknown e^{β_g} — allowing $\sum_j n_{ij} \widehat{\text{den}}_{ij}$ to factor out as a β_g -free constant D_g — $\text{expit}(\beta_g + o_{ij})$ does not factor across distinct o_{ij} : no substitution isolates β_g as a single multiplicative unknown multiplying a β_g -free sum. This already rules out a closed form for the plain score $Y_g = M_g(\beta_g)$ alone; the correction term adds no further obstruction that would change this conclusion, but removes none either.

Practical consequence. Fitting $\hat{\beta}_g$ for the Binomial model therefore requires numerical maximization of $\ell^*(\beta)$ via `brglm2::brglmFit` with `family=binomial`, rather than the two-scalar-sum evaluation of Eq. 16.

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